Geometry-Induced Soft-State Abstractions for Forecasting: Predictive Enrichment with Kernel Affine Hull Machines*

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Abstract

Geometry-induced soft-state abstractions describe an observed history through graded membership in local reference regions. We distinguish these memberships from within-region position and signed reconstruction departures supplied by Kernel Affine Hull Machines (KAHMs). Only the membership block is simplex-valued within each partition. A fixed 32-repetition study of 256 simulated task suites compares four representation-only affine ridge readouts of the same future observations: memberships (G), added local position (P), and KAHM (K) or matched positive-kernel RBF (R) enrichment. In the primary comparison, P selects a common reference bank. P reduces nonlinear loss relative to G by 69.20%, and K reduces P's loss by 10.46%; both establish the prescribed 3% margins within one four-contrast confidence family. K improves R by only 0.03%, leaving its 3% superiority criterion unestablished, while meeting the 5% linear noninferiority criterion against P. Thus 3 of 4 criteria are established. Benefits vary across laws and horizons. We characterize equality and separation of matched KAHM and RBF raw observable spans. Equal spans share exact Koopman-invariance status under a specified deterministic evolution; distinct spans alone imply neither closure nor predictive superiority. Generic bounded-ridge arguments control prediction magnitude without guaranteeing accuracy or rollout stability. Earlier residual and policy studies retain their separate scope, including unsuccessful policy-superiority criteria and an unfavorable descriptive natural-data comparison. The evidence supports local position and conditional reconstruction enrichment as useful predictive observables. It does not establish a general KAHM-specific advantage, membership-only sufficiency, or useful autonomous soft-state dynamics.

1 Introduction

A forecasting abstraction should retain distinctions that matter for the future while making those distinctions accessible to a specified predictor. Geometry-induced soft-state abstractions describe an observed history through graded associations with local reference regions. Memberships can summarize locality while obscuring predictive variation within a region. We study their enrichment with local position and nonlinear reconstruction departures supplied by Kernel Affine Hull Machines (KAHMs) (Kumar et al., 2024). The central question is which observables improve prediction of the same future observation, and what that improvement would establish about a useful state representation.

The construction has three layers. Within each reference partition, nonnegative locality gates sum to one; collecting these vectors gives a membership map Φ_G in a product of simplices. Gate-weighted principal coordinates retain position within the regions. A departure block adds the gate-weighted difference between a nonlinear reconstruction and its affine projection. The enriched raw descriptor ψ_K contains signed coordinates and is not simplex-valued. Neither the membership map nor its enrichment is assumed to be a sufficient or Markov state.

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Our central comparison separates these layers (Section 6). G contains memberships alone; P adds local principal coordinates; K adds KAHM departures to P; and R instead adds matched radial-basis-function (RBF) departures. Every affine ridge head uses only its declared representation to predict the same observed target. There is no retained raw-history forecast or residual target through which missing information could enter. Separate block transformations preserve the G coordinates in all four arms and the P coordinates in both enrichments. This design assesses access through a finite decoder, not unrestricted information: a deterministic representation cannot add information to its input but can expose functions that a restricted predictor can use (Fukumizu et al., 2004; Xu et al., 2020).

The fixed study evaluates 32 independent repetitions of four inherited laws and two noise levels, giving 256 task suites at horizons $1, 4, 12$. Geometry and heads use fitting episodes; separate validation episodes select penalties and, through P, the shared reference bank. Terminal episodes are reserved for evaluation. One four-contrast family asks whether local position improves G, KAHM enrichment improves P, and KAHM improves matched RBF by the prescribed nonlinear margins, while retaining linear noninferiority against P. Independently selected banks address a secondary procedure comparison. They do not replace the primary shared-bank question or its confidence family.

The primary shared-bank results establish three of four criteria. Local position reduces nonlinear loss by 69.20% relative to memberships alone, and KAHM departures reduce the resulting loss by 10.46%; both establish the prescribed 3% margins. KAHM's 0.03% gain over matched RBF does not establish its 3% margin. Its linear loss is higher than P's, but remains within the declared 5% noninferiority criterion. These results support useful geometric enrichment while leaving a general advantage of inverse-Gram reconstruction over the matched positive-kernel alternative unestablished.

KAHM's contribution is identifiable because its matched RBF control shares locality gates, reference histories, projected kernels, and readout architecture. KAHM applies an inverse regularized Gram matrix before normalizing signed reference weights; RBF normalizes positive kernel weights directly. Our analysis characterizes when those different operations yield the same augmented raw observable span and when they permit distinct functions. Normalized KAHM target transfer is a ratio of kernel-ridge predictions, connecting it to rescaled interpolation (Deparis et al., 2014; De Marchi et al., 2017). Span equality allows common coefficients to change; it does not equate corrections around a frozen predictor. Standardization, radial transformations and isotropic ridge penalties further separate raw expressivity from fitted performance.

This distinction also limits the dynamical interpretation. Under a specified deterministic history evolution, equal observable spans have the same exact Koopman-invariance status (Section 4.4). Different spans do not imply that either is closed. Direct horizon-specific forecasts neither require nor establish autonomous coordinate evolution. The companion study (Kumar & Kargaran, 2026a) considers prediction limits for norm-bounded heads and a different, reconstruction-score-normalized simplex map. Its coordinates are not the locality memberships used here. A useful propagation model would require its own common-target and multistep evaluation.

Completed residual-forecast studies provide context, with their original cohorts and comparison families retained in Section 7. The dimension-selected experiment establishes margins over common features and raw-input RBF, together with linear noninferiority, but passes only three of six criteria. Dimension selection strengthens the common representation while reducing the incremental departure benefit in its matched-bank decomposition. Fixed-common follow-ups find useful corrections without establishing the required policy advantage. Generic centered-ridge bounds control correction magnitude; they do not bound total forecasting error.

The subsequent policy study supplies a separate check on practical utility. KAHM augmentation reduces nonlinear policy loss by 0.82% against the base menu and 0.66% against an additional-RBF search control; neither establishes its 3% margin. The descriptive 48-series natural panel is 2.27% worse than the base policy. Those findings motivate isolating representation utility without erasing unsuccessful complete procedures. Fresh repetitions under known laws do not establish transfer to new domains, and the residual-study reproduction adds no independent cohort. The resulting framework separates available observables, their accessibility to a specified decoder, closed coordinate dynamics, and the value of adding a candidate to a forecasting policy.

2 Related work

Soft state representations and local geometry. Soft memberships provide coordinates for describing a state through its association with several reference regions. VAMPnets learn such memberships from time-lagged observations using a variational objective for dynamical modeling (Mardt et al., 2018). In the present construction, locality gates are fitted from input geometry and sum to one separately within each partition. Multiple partitions therefore give a product of simplices. Weighted local coordinates and signed reconstruction departures enrich these memberships with additional observables; the full feature vector is not simplex valued. The memberships are shared by KAHM and the matched positive-kernel control, so the comparison concerns the predictive value of their different enrichments.

Prediction from delay coordinates using local approximation predates the present representations (Farmer & Sidorowich, 1987). Local coordinate coding connects localized representations with approximation of smooth functions (Yu et al., 2009). Local tangent coding adds weighted principal-direction displacements (Yu & Zhang, 2010). Our common block has this gate-and-local-coordinate form, with the locality weights and reference banks specified below. With linear readouts these features are also related to local affine experts (Jacobs et al., 1991). These connections make within-region position an explicit control when asking whether KAHM reconstruction departures change the available functions and improve a specified forecasting procedure.

Kernel reconstruction and feature spaces. KAHMs were introduced for geometrically informed classification, including a differential-privacy application (Kumar et al., 2024), and developed for collaborative learning (Kumar et al., 2025). These works establish the reconstruction construction and geometric properties, including boundedness under their assumptions. The bound here concerns a bounded-feature ridge readout and its residual specializations; the classification results do not establish forecasting utility. The matched positive-kernel reconstruction is a vector-valued kernel smoother that averages reference coordinates with normalized Gaussian weights (Nadaraya, 1964). Rescaled kernel interpolation already divides the interpolant of supplied values by the interpolant of the constant function (Deparis et al., 2014; De Marchi et al., 2017). Proposition 1 records the corresponding identity for regularized KAHM weights, while Propositions 2 and 3 characterize equality and inequality of the corresponding augmented raw feature spans. The central experiment compares representation-only readouts on a bank selected by the common position representation. Its independent-bank context is secondary. Earlier dimension-selected experiments instead compare complete residual procedures with independently selected banks, while their secondary comparisons fix a selected bank. Subsequent shared-baseline comparisons additionally hold the fitted common forecast and bias fixed, isolating the departure correction from jointly refitting the augmented common feature space.

Predictive observables and Koopman models. Reconstruction and forecasting objectives also occur in learned dynamical representations: Lusch et al. (2018) combine reconstruction, latent linearity, and multistep prediction, whereas Liu et al. (2023) optimize a residual Koopman forecaster without the canonical reconstruction loss. Kostić et al. (2022) formulate Koopman regression risk in a reproducing-kernel Hilbert space and analyze reduced-rank operator estimation. These approaches explicitly concern evolution of a chosen observable space. Here separate horizon-specific readouts predict observed future outputs. Those common external targets test predictive utility of the enriched representation without estimating a transition matrix for its memberships or establishing Koopman closure.

The companion study of Kumar & Kargaran (2026a) constructs soft coordinates by normalizing affinities derived from reconstruction scores. That map differs from the center-distance gates used here. It studies successor-coordinate prediction, norm-bounded minimum-risk certificates, fitted-risk upper bounds, and the operator interpretation under exact closure, and includes comparisons on a fixed abstract target. The present study shares KAHM source components and some generator definitions with that work. Its contribution concerns enrichment of geometric memberships and tests against observed-response targets; the coordinate-propagation and certificate experiments remain separate.

Information, regularization, and residual fitting. Supervised dimension reduction studies preservation of the input–target relationship (Fukumizu et al., 2004), and predictive $\mathcal{V}$-information distinguishes information

accessible to a restricted observer (Xu et al., 2020). Our feature-space analysis addresses a finite decoder and does not estimate unrestricted Bayes risk on natural data. Residual fitting under squared loss is an established construction (Bühlmann & Yu, 2003). The correction bound in Proposition 5 follows from the singular values of centered ridge regression and is not specific to KAHM. Nyström approximation changes the available coordinates and computational cost (Williams & Seeger, 2001; Rudi et al., 2015); the raw RBF comparator is therefore evaluated as a complete fitted procedure, not as a presumed equivalent function class.

Evaluation and prediction limits. Model-selection bias (Cawley & Talbot, 2010) and forecast-evaluation pitfalls (Hewamalage et al., 2023) motivate distinct fitting, selection, calibration, and evaluation roles. The supporting natural-data studies use the Monash archive (Godahewa et al., 2021) and are reported separately from the central representation experiment. The risk-certificate framework of Kumar & Kargaran (2026b) develops a block hierarchy of lower risk witnesses and certificates of minimum feasible cost among declared representation classes. Here the algebra compares reconstruction feature spans, the deterministic bound controls fitted readout magnitude, and the paired intervals assess prespecified fitted-procedure comparisons. They do not certify class-wide impossibility or minimum feasible representation cost.

3 Geometry-induced soft-state abstractions and their observables

The representation has three layers: soft membership in reference regions, position within those regions, and nonlinear reconstruction departures. The first layer abstracts a history into graded local assignments. The other layers enrich that abstraction with observables that a finite readout may need. This organization does not assume that the regions are physical regimes, that their memberships are posterior probabilities, or that the resulting descriptor is a sufficient state. An enriched descriptor can also have more coordinates than its input; the term abstraction does not assert dimensionality reduction.

Let (y_t) be a scalar observed series. At forecast origin t, the available history is $x_t = (y_{t-d+1}, \dots, y_t)^\top \in \mathbb{R}^d$, and the target at horizon $h \geq 1$ is y_{t+h}. A forecasting procedure maps the history to a separate prediction for each h in a specified finite horizon set $\mathcal{H}$. The representation and decoder are fitted using a set A of history–target pairs. Distinct episode sets B, C, and D have protocol-specific roles: B selects models; C calibrates the earlier residual procedures but is diagnostic-only in the representation-only representation study; D supplies final evaluation. Sections 6 and 7 specify their respective constructions.

Vectors are columns. The notation $v^\top$ denotes transpose, $\|v\|_2$ the Euclidean norm, and $\|v\|_1$ the sum of absolute coordinates. The matrix I_p is the $p \times p$ identity, and $\mathbf{1}_n$ is the length-n vector of ones; omitted subscripts are determined by the displayed products. Expectations and variances refer to the probability law specified for each statement or experiment. All feature-space statements condition on a fixed fitted reference bank.

3.1 Reference banks and reconstruction maps

Let $z \in \mathbb{R}^d$ be the coordinatewise standardized version of a history x, using means and scales estimated on A. A local reference bank is a collection of fitted models indexed by j. Model j has n_j reference histories stored as the rows of $Q_j \in \mathbb{R}^{n_j \times d}$, center $\mu_j \in \mathbb{R}^d$, and orthonormal principal directions $P_j \in \mathbb{R}^{d \times r_j}$. Here r_j is its local rank and $P_j^\top P_j = I_{r_j}$. Write $q_{ji} \in \mathbb{R}^d$ for reference i, so row i of Q_j is $q_{ji}^\top$. Let S_j be the sample covariance matrix of the projected references $P_j^\top q_{ji}$. For positive rank and positive-definite S_j, the shared Gaussian kernel is

$$\kappa_j(z, z') = \exp\left\{ -\frac{[P_j^\top (z - z')]^\top S_j^{-1} [P_j^\top (z - z')]}{2r_j} \right\}.$$

Define the kernel vector $k_j(z) \in \mathbb{R}^{n_j}$ by $[k_j(z)]_i = \kappa_j(z, q_{ji})$, and the Gram matrix $K_j \in \mathbb{R}^{n_j \times n_j}$ by $[K_j]_{i\ell} = \kappa_j(q_{ji}, q_{j\ell})$. With a positive reconstruction regularizer λ_j, set $C_j = (K_j + \lambda_j I_{n_j})^{-1}$. The Gaussian Gram matrix and its regularized inverse C_j are symmetric. The regularizer is fitted from A by the reference-reconstruction rule specified in Appendix C.

The KAHM and matched positive-kernel reconstructions are

$$A_{K,j}(z) = \frac{Q_j^\top C_j k_j(z)}{\mathbf{1}^\top C_j k_j(z)}, \qquad A_{R,j}(z) = \frac{Q_j^\top k_j(z)}{\mathbf{1}^\top k_j(z)}. \tag{1}$$

Both take values in $\mathbb{R}^d$. The first is defined when its signed denominator is nonzero; the second has a positive denominator for finite inputs. The matched affine reconstruction is $A_{A,j}(z) = \mu_j + P_j P_j^\top (z - \mu_j)$.

Local dimension. For a supplied cap $c \geq 1$, the native construction initializes the local dimension at

$$r_j^{\text{init}} = \min\{c, d, n_j - 1\}.$$

The default cap is $c = 20$ (Kumar et al., 2025, Appendix B). The stored rank r_j can be smaller because the native PCA retains only positive-variance directions and reduces the requested dimension when a projected reference coordinate has insufficient range. The range rule stops at requested rank one; it does not impose the range threshold on every possible rank-one model. Appendix C gives the exact numerical rules and the explicit rank-zero wrapper. There is no additional $d - 1$ cap in this native policy. In particular, full ambient rank is permitted, and a cap of twenty does not imply twenty stored directions when $d < 20$ or $n_j \leq 20$.

3.2 Soft memberships and within-state observables

Models are grouped into L partitions. If $\mathcal{J}_b$ indexes the models in partition b, their shared locality weights are

$$g_j(z) = \frac{\exp\{-\|z - \mu_j\|_2^2/(2s_j)\}}{\sum_{\ell \in \mathcal{J}_b} \exp\{-\|z - \mu_\ell\|_2^2/(2s_\ell)\}}, \qquad j \in \mathcal{J}_b, \tag{2}$$

where $s_j > 0$ is the fitted local squared-radius scale. These weights, called gates, sum to one within each partition and do not depend on the reconstruction being compared. Define the probability simplex $\Delta_q = \{s \in \mathbb{R}^q : s_i \geq 0,\ \mathbf{1}^\top s = 1\}$ and the partition-wise membership map

$$\Phi_G^{(b)}(z) = (g_j(z))_{j \in \mathcal{J}_b} \in \Delta_{|\mathcal{J}_b|}, \qquad \Phi_G(z) = (\Phi_G^{(1)}(z), \ldots, \Phi_G^{(L)}(z)) \in \prod_{b=1}^{L} \Delta_{|\mathcal{J}_b|}. \tag{3}$$

We identify this product with its concatenated vector in a fixed order. There is no additional normalization across partitions. Thus each partition gives a soft assignment to its reference regions, and the complete gate vector has coordinate sum L. These assignments are the geometry-induced soft-state abstraction used by the present construction.

The common feature vector $\psi_0(z)$ contains all gates $g_j(z)$ and all weighted principal-coordinate vectors $g_j(z)P_j^\top(z - \mu_j)$. The gates describe relative locality; the weighted coordinates retain position within the regions. Equal memberships need not imply equal common features. The common vector therefore already enriches the membership-only abstraction, and a comparison against it is not a test against memberships alone.

3.3 KAHM enrichment of the shared abstraction

Define the reconstruction departures by

$$\begin{aligned} d_K(z) &= \text{concat}_j\big[g_j(z)\{A_{K,j}(z) - A_{A,j}(z)\}\big], \\ d_R(z) &= \text{concat}_j\big[g_j(z)\{A_{R,j}(z) - A_{A,j}(z)\}\big]. \end{aligned} \tag{4}$$

The operator concat stacks its column-vector arguments in a fixed order. Each departure retains all history coordinates, including components outside the local principal subspace. Substituting the affine map for the nonlinear reconstruction makes the departure identically zero. The common-feature procedure is therefore the nested control without reconstruction departures. Within a fixed bank, the RBF departure is matched to the KAHM departure in references, kernels, gates, and feature dimension.

The augmented raw descriptors are

$$\psi_K(z) = \mathrm{concat}(\psi_0(z), d_K(z)), \qquad \psi_R(z) = \mathrm{concat}(\psi_0(z), d_R(z)). \tag{5}$$

Local coordinates and departures can be signed, so these descriptors are not simplex-valued. KAHM changes the observables attached to the shared soft assignments through its inverse-Gram reconstruction; it does not change the gate map in a matched-bank comparison. In particular, the signed coefficients inside $A_{K,j}$ are distinct from the nonnegative gates in Equation (2).

The affine control is nested at the level of the raw feature construction. The comparison with positive-kernel RBF reconstruction additionally holds fixed the number and locations of references, the projected kernel, the locality weights, and the decoder architecture. The difference between the two raw departure blocks is therefore the use of C_j in the normalized reconstruction. This matching does not assert equality of fitted predictions, and it does not compare KAHM with every possible kernel or nonlinear decoder. When complete procedures select their caps independently, their selected banks and common-feature dimensions can differ. The earlier dimension study distinguishes its primary complete-procedure comparisons from secondary comparisons holding the KAHM-selected bank fixed. The representation-only study instead makes the P-selected shared bank primary and independently selected banks secondary.

Relation to reconstruction-score abstractions. An alternative geometry-induced soft-state map normalizes affinities derived from scalar KAHM reconstruction scores (Kumar & Kargaran, 2026a). That operation makes the scores themselves determine membership. The present map Φ_G instead uses distances to fitted reference centers, and the enrichment retains vector-valued departures from local affine reconstruction. The two maps can retain different information. The experiments here evaluate the latter construction on common future observations; they do not evaluate successor-coordinate prediction for the former score-based map.

Usefulness is relative to an external prediction task. For each horizon, all compared procedures predict the same observed y_{t+h} under the same loss and evaluation sample. A useful enrichment must improve that task through the declared fitting and selection procedure. Accurate reference assignment or small reconstruction error alone does not establish this property. The retained raw-history or common forecast can also contribute information outside a departure block. We therefore assess complete forecasts and matched corrections, without identifying their accuracy with sufficiency of Φ_G. Section 4.4 separately states what would be required to propagate an observable representation through time.

3.4 Defined evaluation of signed normalizations

To guard against cancellation in the signed KAHM normalization, define

$$a_j(z) = C_j k_j(z), \qquad \rho_j(z) = \frac{|\mathbf{1}^\top a_j(z)|}{\|a_j(z)\|_1} \quad \text{when } \|a_j(z)\|_1 > 0. \tag{6}$$

The ratio is invariant to a common positive rescaling of the query kernel. When $\rho_j(z) \le 10^{-10}$, or a required quantity is undefined or nonfinite, the local reconstruction is replaced by its affine map, making that raw departure zero. The same finite-value checks apply to the positive RBF map. All affected rows are recorded. This fallback gives a defined feature vector where the original ratio cannot be evaluated reliably; it does not establish continuity across the fallback boundary.

A zero raw departure under fallback need not remain zero after A-fitted feature centering. The fallback defines the input to the subsequent transformation; the norm bound in Section 5 applies to that transformed input. The algebraic propositions in Section 4 concern the exact ratios on their stated domains, rather than the piecewise numerical fallback procedure.

4 Predictive observable spaces and their dynamical interpretation

The enriched abstraction raises three questions: how normalized reconstruction relates to established kernel prediction, whether its coordinates give a linear readout functions unavailable from the matched reconstruction,

and what the resulting observable space can imply about exact dynamical closure. We answer these questions for fixed reference geometry. The results concern exact functions before feature standardization, radial transformation, or penalized fitting.

Consider one local model from Section 3 and suppress its index. Write M for its number of references, $Q \in \mathbb{R}^{M\times d}$ for their row-stored coordinates, and $C = (K + \lambda I_M)^{-1}$ for its regularized inverse, with $\lambda > 0$. The matrix K is the symmetric positive-semidefinite Gaussian Gram matrix. The vector $\mathbf{1}_M$ contains M ones, and $k(z) \in \mathbb{R}^M$ contains the kernel similarities of a query z to the references. Let A_K, A_R, and A_A denote the KAHM, positive-kernel RBF, and affine reconstructions in Equation (1).

4.1 The meaning of normalized target transfer

Proposition 1 (Normalization and target transfer)**.** *For a query with $s(z) = \mathbf{1}_M^\top Ck(z) \neq 0$, define $\omega(z) = Ck(z)/s(z)$. Then $\mathbf{1}_M^\top \omega(z) = 1$, and for every vector of reference targets $y \in \mathbb{R}^M$,*

$$\omega(z)^\top y = \frac{k(z)^\top (K + \lambda I_M)^{-1} y}{k(z)^\top (K + \lambda I_M)^{-1} \mathbf{1}_M}. \tag{7}$$

This is the ratio of two kernel-ridge predictions with regularization matrix $K + \lambda I_M$ and no additional intercept. For an affine scalar function $f(q) = q^\top b + c$, where $b \in \mathbb{R}^d$ and $c \in \mathbb{R}$, its reference values $f(Q) = Qb + c\mathbf{1}_M$ satisfy

$$\omega(z)^\top f(Q) - f(z) = \{A_K(z) - z\}^\top b.$$

The proof uses symmetry of C and the fact that the normalized weights sum to one. The numerator in Equation (7) fits the supplied targets, while the denominator fits the constant target. Because the denominator varies with the query, the ratio generally differs from the ordinary unnormalized kernel-ridge predictor. The affine identity also specifies the effect of reconstruction error on an affine target: only its component in the target direction b enters. The complete proof is in Appendix A.

The quotient normalization has an established interpolation antecedent. When K is invertible, setting $\lambda = 0$ in the quotient gives the rescaled kernel interpolant of De Marchi et al. (2017), wherever its denominator is nonzero. Proposition 1 interprets the existing KAHM weights with positive regularization; it does not introduce a new normalization principle.

4.2 Equal spans under reference symmetry

For a fixed common gate $g(z)$, define $d_K(z) = g(z)\{A_K(z) - A_A(z)\}$ and $d_R(z) = g(z)\{A_R(z) - A_A(z)\}$. Let $\mathcal{S}_0$ be the linear span of the scalar gate and the coordinates of $g(z)P^\top(z - \mu)$, where μ and P are the reference center and principal directions. The coordinates of $g(z)A_A(z) = g(z)\mu + P\{g(z)P^\top(z - \mu)\}$ belong to $\mathcal{S}_0$. Adjoining a vector-valued feature to a span means adjoining each of its scalar coordinate functions. A common intercept may be adjoined to both spans without changing the equality below.

Because gA_A already belongs coordinatewise to $\mathcal{S}_0$, adjoining a departure gives the same raw span as adjoining the corresponding gated reconstruction. Subtracting A_A specifies a parameterization; the span comparison concerns the additional functions supplied by the nonlinear map.

Proposition 2 (A sufficient condition for equal feature spans)**.** *Suppose $K\mathbf{1}_M = \kappa\mathbf{1}_M$ for a scalar κ, and C leaves $\mathcal{V} = \operatorname{span}\{\mathbf{1}_M, Q_{:1}, \ldots, Q_{:d}\}$ invariant. Here $Q_{:i}$ is column i of Q, and invariance means $C\mathcal{V} \subseteq \mathcal{V}$. Then*

$$\operatorname{span}(\mathcal{S}_0, d_K) = \operatorname{span}(\mathcal{S}_0, d_R).$$

More explicitly, there are $B \in \mathbb{R}^{d\times d}$ and $c \in \mathbb{R}^d$ such that, with $a = (\kappa + \lambda)^{-1} > 0$,

$$C\mathbf{1}_M = a\mathbf{1}_M, \qquad CQ = QB + \mathbf{1}_M c^\top, \qquad A_K(z) = \{B^\top A_R(z) + c\}/a.$$

Invariance supplies the displayed representation of CQ. Substitution in the normalized reconstruction expresses each KAHM departure using RBF departures and the common affine features. The restriction of C

to the finite-dimensional space $\mathcal{V}$ is invertible onto that space, which gives the reverse inclusion. This argument also covers redundant reference-coordinate columns; it does not require B to be invertible. Appendix A gives the full proof.

A two-reference example makes the role of the common coordinates explicit. Take scalar references $(-1, 1)$, $g = 1$, $P = 1$, and $A_A(z) = z$. If ρ is the off-diagonal Gram entry, then

$$A_K = c_\rho A_R, \qquad d_K = c_\rho d_R + (c_\rho - 1)z, \qquad c_\rho = \frac{1+\rho+\lambda}{1-\rho+\lambda} > 0.$$

The departures differ, but adjoining the coordinate z makes their augmented spans equal. Thus a nonlinear reconstruction formula alone does not establish an additional raw basis.

A fixed common predictor changes the comparison. The augmented spans above allow the common coefficients to vary. If a common forecast f_{base} is fixed, the relevant affine function classes can instead be

$$\mathcal{F}_K = f_{\text{base}} + \operatorname{span}\{d_K\}, \qquad \mathcal{F}_R = f_{\text{base}} + \operatorname{span}\{d_R\}. \tag{8}$$

For the two-reference example on the whole real line, $0 < \rho < 1$ implies $c_\rho \neq 1$, and z and d_R are linearly independent. Consequently $\mathcal{F}_K \neq \mathcal{F}_R$, although the jointly augmented spans are equal. Neither correction class contains the other. Appendix A.3 proves these statements. Centering preserves the displayed linear identity, but the same argument for centered corrections requires independence after centering on the query domain. Separate nonlinear radial transformations require their own comparison. This example clarifies a restriction imposed by freezing the common predictor; it supplies no ordering of forecasting risks or attribution of the experimental differences to that restriction.

4.3 An irregular reference bank gives a distinct basis

Proposition 3 (Irregular scalar references)**.** *Let $q_1 < q_2 < q_3$ satisfy $q_1 + q_3 \neq 2q_2$, and let $k_i(z) = \exp\{-t(z - q_i)^2/2\}$ for a precision $t > 0$. Set $K_{ij} = \exp\{-t(q_i - q_j)^2/2\}$ and $C = (K + \lambda I_3)^{-1}$ for $\lambda > 0$. Suppose every entry of $C\mathbf{1}_3$ is positive. With $Q = (q_1, q_2, q_3)^\top$, $g = 1$, and $A_A(z) = z$,*

$$A_K \notin \operatorname{span}(1, z, A_R), \qquad d_K \notin \operatorname{span}(1, z, d_R).$$

These spans consist of scalar functions on the whole real line.

The six products of kernel functions have distinct exponential factors because their exponents contain the six distinct sums $q_i + q_j$ for $i \leq j$. Their linear independence rules out an affine relation between A_K and A_R: such a relation would force K to have constant row sums, whereas its middle row has a strictly larger sum than either outer row. Positivity of $C\mathbf{1}_3$ makes both reconstructions bounded, excluding an additional nonzero multiple of z. Appendix A provides every step of this argument.

For the explicit references $(-1, 0, 2)$, the sample-variance precision is $t = 3/7$. The reference-based regularization rule gives $\lambda = 4.498274743339644$, for which positivity of $C\mathbf{1}_3$ follows from the matrix bound in the appendix. This supplies a concrete example satisfying the proposition. It does not assert distinctness for every higher-dimensional bank fitted in the forecasting experiment.

Together, the propositions delimit the algebraic contribution. Some reference geometries yield equal augmented spans, while an irregular geometry can supply a distinct function. Neither conclusion determines forecasting accuracy. A general change of feature coordinates changes an isotropic ridge penalty, and separate nonlinear radial transformations can remove a raw linear span relation. The empirical comparison therefore evaluates the complete feature transformation and fitted decoder as well as the reconstruction map.

4.4 Soft state coordinates and dynamical observable spaces

For this discussion, write all features as functions of the history x, absorbing their fixed preprocessing. The map Φ_G in Equation (3) describes locality to reference regions in a product of simplices. The enriched raw descriptors in Equation (5) supply additional, potentially signed observables. Neither construction is assumed to be a sufficient Markov state. The following analysis asks when a specified observable family can evolve autonomously and how that requirement relates to prediction of an external output.

Direct readouts and propagated coordinates. For a fixed soft representation Φ, a dynamical forecasting model can combine one propagation matrix with an output decoder:

$$\widehat{y}_h^{\mathrm{prop}}(x) = d_{\mathrm{out}}\big((B^\top)^h \Phi(x)\big). \tag{9}$$

If $d_{\mathrm{out}}(s) = v^\top s$, then $\widehat{y}_h^{\mathrm{prop}}(x) = (B^h v)^\top \Phi(x)$. The horizon-specific coefficient vectors therefore share the recursion $v_{h+1} = Bv_h$. Independent direct linear readouts $v_h^\top \Phi(x)$ do not impose that relation. Propagation can provide useful structure, but it also requires an adequate evolution model for the coordinates. The central experiment fits representation-only direct readouts for a finite horizon set. Earlier studies additionally use retained baselines and calibrated corrections. Neither construction's accuracy establishes a matrix that propagates the feature vector. In the earlier architecture, a history-dependent baseline can retain predictive information absent from the departure features themselves.

A function-space criterion for exact evolution. Let $F : \mathcal{X} \to \mathcal{X}$ be a fixed deterministic evolution on a specified history-state domain. This is an additional assumption for the following statement; finite observed histories need not define such a deterministic state. For any finite raw feature vector ψ, set

$$\eta_\psi(x) = (1, \psi(x)^\top)^\top, \qquad \mathcal{V}_\psi = \mathrm{span}\{\text{coordinates of } \eta_\psi\}, \qquad K_F g = g \circ F. \tag{10}$$

These are spaces of functions on $\mathcal{X}$, without an assumed inner product or a choice of matrix-fitting algorithm.

Proposition 4 (Raw feature spans and exact dynamical closure)**.** *The space $\mathcal{V}_\psi$ is invariant under K_F if and only if there exists a real square matrix T_ψ such that*

$$\eta_\psi(F(x)) = T_\psi^\top \eta_\psi(x) \quad \textit{for every } x \in \mathcal{X}.$$

Consequently, if the augmented raw KAHM and matched RBF feature spaces satisfy $\mathcal{V}_{\psi_K} = \mathcal{V}_{\psi_R}$, they have the same exact invariance status under this F. When invariant, their induced operators on that common function space are identical. Matrix representations in nonredundant bases are similar; full matrices on redundant coefficient vectors need not faithfully represent the operator's spectrum.

Proof. If the space is invariant, each coordinate of $\eta_\psi \circ F$ is a linear combination of the coordinates of η_ψ; collecting these coefficients gives T_ψ. Conversely, the displayed identity gives $K_F(c^\top \eta_\psi) = (T_\psi c)^\top \eta_\psi \in \mathcal{V}_\psi$ for every coefficient vector c. Equal function spaces have the same image under composition with F, proving the invariance equivalence and equality of the restricted operators. Two bases of that space are related by an invertible change of coordinates, so their operator matrices are similar. With redundant coordinates, let $\mathcal{N}_\psi = \{c : c^\top \eta_\psi \equiv 0\}$. The identity implies $T_\psi \mathcal{N}_\psi \subseteq \mathcal{N}_\psi$. The faithful action is on the quotient by this null space; eigenvectors contained in it represent the zero function. □

Proposition 4 is the standard observable-space implication of exact closure, applied to the raw span comparisons here (Kumar & Kargaran, 2026a). Under the hypotheses of Proposition 2, changing between matched KAHM and RBF coordinates cannot change exact invariance of their common augmented span. Proposition 3 instead permits different candidate observable spaces; it does not show that either is invariant or more predictive. The assertion concerns jointly variable augmented spans, not the restricted correction classes with a frozen common forecast in Equation (8).

Prediction targets and the limits of closure claims. An adequate propagated representation must also support the output of interest. Comparing abstractions through their respective successor coordinates changes the target along with the representation; nearly constant coordinates can be easy to forecast while retaining little useful state information. The forecasting experiments instead keep the future observation y_{t+h} fixed across compared procedures. They test access to that target through finite readouts, not internal closure of a soft-coordinate map. A proposed propagation model requires its own common-target and rollout evaluation.

Exact invariance concerns pointwise functions on the stated domain. Zero population squared error implies only almost-sure equality under its evaluation law, and a small empirical residual establishes neither statement.

Under stochastic transitions, an equation $\mathbb{E}[\Phi(x_{t+1}) \mid x_t = x] = B^\top \Phi(x)$ concerns the conditional mean. Iteration additionally requires suitable Markov and time-homogeneity assumptions; realized successors can retain random variation. Correction-magnitude bounds do not establish these properties or control the amplification of errors through matrix powers.

Finally, exact raw-space invariance is unchanged by a change of basis, whereas isotropic ridge penalties and matrix-norm limits generally are not. Separate nonlinear radial transforms can change the observable space itself. The minimum-risk certificates for fixed simplex-valued coordinates in Kumar & Kargaran (2026a) additionally specify a successor-coordinate target, predictor norm and independent evaluation units. Applying such certificates to these enriched signed features or raw-target residual heads requires a separately justified formulation. The present span results therefore characterize available observables; the experiments determine their conditional forecasting value under the declared transformations, readouts and selection procedures.

5 Direct forecasting readouts of the soft-state hierarchy

The representation comparison uses affine ridge heads that predict the observed future directly from their declared coordinates. For a fixed fitted bank, G supplies memberships, P adds local position, and K and R add their respective departures. Each horizon has its own head; no transition matrix is fitted or iterated. This section specifies the transformations and readouts. The four-arm comparison and its primary selection context are given in Section 6.

5.1 Separate A-fitted block transformations

Let $b_G = \Phi_G$, b_P be the gate-weighted principal coordinates, and $b_K = d_K$, $b_R = d_R$ be the raw departure blocks. Each block is centered coordinatewise by its A mean and divided by its A standard deviation, floored at 10^{-6}; a tilde denotes this transformation. These parameters remain fixed for B, C and D. For a nonempty block of width p, define

$$T_R(v) = \frac{v}{\sqrt{1 + \|v\|_2^2/R^2}}, \qquad R = \sqrt{p}, \qquad v \in \mathbb{R}^p. \tag{11}$$

Set $z_\ell = T_{\sqrt{p_\ell}}(\widetilde{b}_\ell)$ for $\ell \in \{G, P, K, R\}$. An empty position block remains empty, with radius zero. The radial map has norm at most R and is nonexpansive in its argument; Appendix A gives the elementary proof.

The inputs are $\chi_G = z_G$, $\chi_P = [z_G, z_P]$, $\chi_K^{\mathrm{sep}} = [z_G, z_P, z_K]$, and $\chi_R^{\mathrm{sep}} = [z_G, z_P, z_R]$. The superscript distinguishes these inputs from the earlier joint-common transformation in Appendix B. No further transformation is applied to a concatenated input. Its G block is identical in all four arms, and its P prefix is identical in both enrichments. Centering and radial transformation generally remove the simplex constraint from the fitted input even though the raw gates remain simplex-valued within each partition.

For J local models with ranks r_j and histories in $\mathbb{R}^d$, $p_G = J$, $p_P = \sum_j r_j$, and $p_K = p_R = Jd$. Write χ_m generically for arm m's input, including the superscripted K and R vectors. It satisfies $\|\chi_m(x)\|_2 \le R_m$, where R_m^2 is the sum of its block widths. These bounds use the actual fitted dimensions, including zero ranks. Equal K/R width does not equate their function spaces or conditioning.

5.2 Direct affine ridge heads

For each arm, candidate bank and horizon h, the fitted head is

$$\begin{aligned} (\widehat{b}_{m,h}, \widehat{w}_{m,h}) &= \operatorname*{arg\,min}_{b,w} \left[\frac{1}{n_A} \sum_{i \in A} \{y_{i,h} - b - \chi_m(x_i)^\top w\}^2 + \alpha_{m,h} \|w\|_2^2 \right], \\ \widehat{f}_{m,h}(x) &= \widehat{b}_{m,h} + \chi_m(x)^\top \widehat{w}_{m,h}. \end{aligned} \tag{12}$$

The intercept is unpenalized. The targets are observations, not residuals, and there is no retained baseline, correction multiplier or subsequent calibration. B selects the penalty from $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$ separately for each arm and horizon. The primary bank is selected by P's mean variance-normalized B

loss across horizons; each arm retains its own selected penalties on that bank. Geometry, transforms and coefficients remain fitted on A. C supplies diagnostics after selection and changes no fitted quantity; D evaluates the frozen forecasts. Secondary independent bank selection is kept separate from that primary context.

Direct-readout corollary of the centered-ridge bound. Let $\bar{y}_h$ and $\sigma_{y,h}$ be the A target mean and centered root mean square, and let $\bar{\chi}_m$ be the A feature mean. Proposition 5 in Appendix B, with $e_{ih} = y_{i,h}$ and $\gamma_h = 1$, gives

$$\begin{aligned} \|\widehat{w}_{m,h}\|_2 &\le \frac{\sigma_{y,h}}{2\sqrt{\alpha_{m,h}}}, \\ |\widehat{f}_{m,h}(x)| &\le |\bar{y}_h| + (R_m + \|\bar{\chi}_m\|_2)\frac{\sigma_{y,h}}{2\sqrt{\alpha_{m,h}}}. \end{aligned} \tag{13}$$

This follows by eliminating the intercept and bounding the centered ridge coefficient operator. It is a generic magnitude bound, not a new KAHM-specific guarantee. It supplies no bound on prediction error, informativeness, signed-reconstruction sensitivity, or recursive rollout stability.

The preceding studies use two different residual architectures: joint common/departure fitting around a retained raw-history baseline, and centered departure slopes around a frozen common forecast and shared bias. Their calibration, intercept treatment, and complete bounds remain in Appendix B. Those architectures can access information through the retained predictor that is absent from the correction features; their historical gains therefore answer a different question from the representation-only comparison.

6 Predictive utility of the soft-state hierarchy

The experiment separates membership, local position, and reconstruction enrichment. Every readout predicts the same future observation using only its declared representation. This tests which geometric observables a finite decoder can use without a retained raw-history forecast supplying information outside those observables. The comparison concerns fitted predictive accessibility; it does not establish unrestricted information or Markov sufficiency.

6.1 Representations and shared transformations

Write $b_G = \Phi_G$ for the concatenated gates and b_P for the gate-weighted principal coordinates. Along with d_K and d_R, these give four raw blocks. Each block is centered and scaled on A alone, with scale floor 10^{-6}, and separately transformed by the radial map in Equation (11). Denote the resulting blocks by z_G, z_P, z_K, z_R. An empty position block remains empty. The four readout inputs are

$$\chi_G = z_G, \qquad \chi_P = \mathrm{concat}(z_G, z_P), \qquad \chi_K^{\mathrm{sep}} = \mathrm{concat}(z_G, z_P, z_K), \qquad \chi_R^{\mathrm{sep}} = \mathrm{concat}(z_G, z_P, z_R). \tag{14}$$

We refer to these procedures as G, P, K, and R. The superscript sep distinguishes the separately transformed gate and position blocks from the earlier joint-common transformation in Appendix B. The raw observables are unchanged. The gate block is identical in all four inputs, and the position block is identical in P, K, and R. Appending an observable block therefore does not change the existing coordinates. K and R have the same nominal width and readout architecture; this matching does not equate effective function spaces, conditioning, or computational cost.

For each cap and horizon, an affine ridge head is fitted directly to the observed A targets, with an unpenalized intercept. B selects a penalty from $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$ independently for each arm and horizon. There is no residual target, retained baseline, correction multiplier, or calibration. C predictions are recorded as diagnostics after all choices are fixed and do not affect fitting or selection.

6.2 Validation-selected geometry and fresh evaluation

Each candidate bank is fitted from A using the native construction in Section 3. All four arms share its reference histories, partitions, gates, projected kernels, and local ranks. Candidate caps are $1, 2, 4, 8, 12$ at

history width twelve; cap twenty is an alias of twelve and does not receive an additional search slot. The native variance and range reductions remain in force. Input histories use A-fitted standardization, with no additional scaler search.

In the primary context, P selects one cap by mean B loss across horizons, using the corresponding A-target population variance for normalization and its previously selected horizon penalties. Exact ties choose the first cap in ascending order. G, K, and R then use that same bank with their own B-selected penalties. This estimates enrichment of a validation-selected common representation. A secondary context allows each arm to select its own cap from the same candidate fits; it measures a complete procedure and cannot replace a primary comparison.

The fixed design specifies 32 independent repetitions of four inherited laws and two added-noise levels, giving 256 task suites. Histories have width twelve and horizons are $1, 4, 12$. Independent A/B/C/D sets contain 64/32/32/128 episodes of length seventy-two. A/B/C retain at most 512/256/256 rows; D retains all $1,664$ origins. The observation laws, noise conventions, window indexing, and paired-noise construction are those specified in Appendix C. These fresh repetitions extend evidence under known laws; they do not introduce new domains. Earlier outcomes informed the design; the protocol excludes pooling earlier observations into its inferential units.

Source, configuration, analysis, and the repetition budget were frozen and the scientific source was published before production data generation. The complete A/B fitted-state roster was sealed before D generation, and all D forecasts were sealed before terminal scoring. Source hashes and phase records document this ordering; they do not constitute independent oversight or cryptographic target isolation. The pipeline retains candidate choices, portable fitted states, forecasts, numerical recoveries, and complete fitting-and-selection costs.

6.3 Primary comparison family

Task losses average variance-normalized squared error over origins and horizons. Within each repetition, nonlinear losses average the three nonlinear laws and both noise levels equally; the linear control averages its two noise levels. The independent units are the 32 repetition aggregates, not overlapping windows. There is one family of four primary claims: nonlinear P versus G, K versus P, and K versus R at a 3% superiority margin, and linear K versus P at a 5% noninferiority margin. These are inherited research thresholds rather than deployment-derived costs.

For each claim, let $d_r = L_{a,r} - cL_{b,r}$, where a and b are the candidate and reference and $c = 0.97$ for superiority or 1.05 for noninferiority. With $n = 32$, the one-sided upper endpoint is

$$U = \bar{d} + t_{n-1,1-0.05/4}\, s_d/\sqrt{n}. \tag{15}$$

Here $t_{\nu,p}$ is the p quantile of a Student distribution with ν degrees of freedom. A criterion is established only if $U < 0$. This Bonferroni family uses the paired-t model for independent repetition aggregates; its finite sample accuracy depends on that approximation. Percentage gains are $100\{1 - \overline{L}_a/\overline{L}_b\}$. Independent-cap, law, noise, horizon, and resource analyses are secondary and descriptive. An unestablished margin is not evidence of equivalence.

6.4 Results and their scope

Table 1 reports all four primary comparisons. P reduces nonlinear loss relative to G by 69.20%, and K reduces loss relative to P by 10.46%. Both simultaneous upper endpoints are below zero, establishing the respective 3% margins. The mean K/R difference is much smaller: K improves R by only 0.03%, with a positive margin-adjusted upper endpoint. The KAHM-specific 3% superiority criterion is not established. On the linear control, K's loss exceeds P's by 1.11%, while the upper endpoint for the 5% noninferiority criterion remains negative. The complete family therefore establishes 3 of 4 criteria, supporting position and KAHM enrichment relative to their nested inputs under this procedure, without establishing superiority over matched RBF enrichment.

Table 2 shows that independent cap selection reduces nonlinear K and R losses to 0.0267546 and 0.0267507 respectively. K's descriptive gain over P becomes 12.14%, while its K/R gain becomes −0.01%. These point

Table 1: Complete primary family for the P-selected shared bank. Gain is $100(1 - \overline{L}_a/\overline{L}_b)$; negative gain denotes higher candidate loss. The margin-adjusted difference is $d = L_a - cL_b$, with $c = 0.97$ for superiority and 1.05 for noninferiority. A criterion is established only when its simultaneous one-sided upper endpoint is below zero. The independent units are the 32 repetition aggregates.

Population	Contrast	Criterion	Gain (%)	$\bar{d}$	Upper bound	Met
Nonlinear	P vs G	3% sup.	69.20	−0.06545	−0.06253	Yes
Nonlinear	K vs P	3% sup.	10.46	−0.002271	−0.001987	Yes
Nonlinear	K vs R	3% sup.	0.03	0.0008103	0.000981	No
Linear	K vs P	5% NI	−1.11	−0.03542	−0.03134	Yes

Table 2: Mean variance-normalized terminal loss and standard deviation across repetitions. The shared bank is selected by P. Independent per-arm cap selection is a secondary procedure comparison; its outcomes cannot replace a primary criterion. Standard deviations describe repetitions and are not confidence intervals.

Cap rule	Arm	Nonlinear mean	SD	Linear mean	SD
Shared	G	0.0988703	0.00788637	0.939042	0.0484704
Shared	P	0.0304507	0.00150099	0.910921	0.0513907
Shared	K	0.027266	0.00123317	0.921045	0.0526607
Shared	R	0.027274	0.00127755	0.915866	0.0512334
Independent	G	0.0988703	0.00788637	0.939042	0.0484704
Independent	P	0.0304507	0.00150099	0.910921	0.0513907
Independent	K	0.0267546	0.00120524	0.914936	0.0511589
Independent	R	0.0267507	0.00129299	0.91589	0.0521994

estimates do not establish equivalence or a primary advantage under a different selection rule. G's forecasts are identical across caps in this construction: the shared seed and references leave the center-distance gates unchanged. Its independent choice of cap one is consequently a tie-breaking outcome, not evidence that rank one best captures the dynamics. P's independently chosen cap is the primary cap by definition.

The aggregate gains are heterogeneous (Figure 1). K improves P on both oscillator laws at both noise levels, but worsens P on nonlinear autoregression at both levels. It improves matched R in three of the six nonlinear law/noise cells and worsens it in the other three. By horizon, nonlinear K/P loss is higher at $h = 1$, nearly unchanged at $h = 4$, and lower at $h = 12$. These are descriptive decompositions of the complete results, not additional confirmed subpopulation claims. They show why a positive aggregate enrichment result does not justify universal deployment of the departure block.

The Supplementary Studies reports all law/noise and horizon losses, both cap-selection contexts, dimensions, and resource accounting. Selected G, P, and K/R widths are 14, 28–182, and 196–350, respectively, for history width twelve. These descriptors are therefore not a compression claim. Complete fitting and selection uses 0.26 summed worker wall hours and 0.26 CPU hours; D prediction adds 671.84 summed worker wall seconds. Peak measured worker memory is 156.99 MiB. These costs include all candidates and reused geometry. Summed worker time is not elapsed parallel runtime, and the measurements do not identify isolated K versus R deployment costs. Preparation, audit and document production are outside these fitting-and-prediction measurements.

Every planned task and primary contrast is retained. The source, budget, menu and comparison family were unchanged after publication; unfavorable comparisons did not trigger additional repetitions. The experiment establishes finite direct-readout utility under the specified laws and allocation. It does not test autonomous coordinate propagation, Markov sufficiency, or a generally preferable forecasting policy.

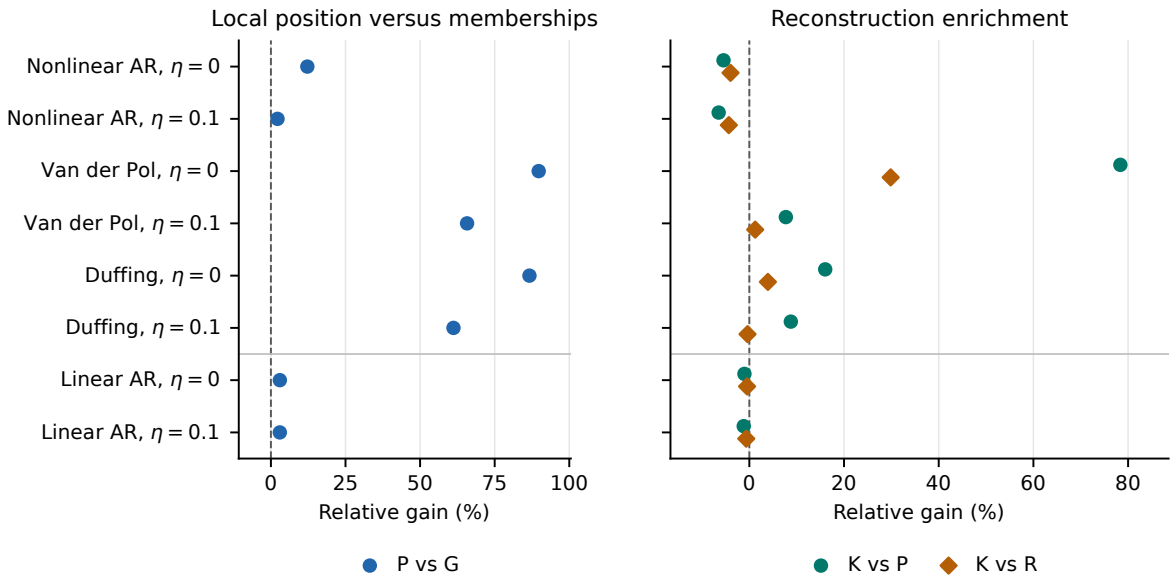


Figure 1: Descriptive shared-bank gains by observation law and added noise. Positive values favor the candidate. Each point uses all 32 repetitions and averages the three horizons; the condition displays have no separate confirmatory status. The complete primary confidence family is in Table 1.

7 Context from preceding forecasting studies

The preceding studies evaluate complete forecasting procedures and residual corrections. They provide context for the representation ablation without supplying additional observations to its primary analysis. Each retains its original cohort, selection rules, comparison family, and numerical outcomes. Appendix C gives the dimension-study protocol and complete results; the *Supplementary Studies* gives the later protocols and detailed tables.

7.1 Validation-selected local dimension

The earlier dimension experiment compares ten procedures on four laws (nonlinear autoregression, Van der Pol, Duffing, and linear AR(2)), two added-noise levels, and 24 independent repetitions: 192 tasks. Histories have width twelve and horizons $1, 4, 12$. Independent episode sets A/B/C/D contain 64/32/32/128 episodes: A fits scalers, banks and heads; B selects hyperparameters; C calibrates corrections and selects the non-KAHM procedure; D evaluates frozen forecasts. A/B/C retain at most 512/256/256 histories, and D uses all 1,664 origins. Source and analysis were frozen locally before generation; predictions were sealed before D-target release. This was not public preregistration.

Common, KAHM, and RBF procedures select their native cap independently from $1, 2, 4, 8, 12, 20$. Primary comparisons therefore need not share a bank; secondary matched-bank comparisons hold the KAHM-selected bank fixed. All non-ridge candidates are calibrated toward the same raw-ridge baseline. The non-KAHM policy selects among ridge, raw RBF, boosting, persistence, common features, and bounded or unbounded RBF departures. Its comparison assesses a complete procedure, not matched feature capacity. Native caps twelve and twenty coincide at the tested history width; ascending ties select twelve. Native dimension permits full ambient rank. The historical fixed-cap-two control retains its specified $\min\{2, d-1, n_j-1\}$ restriction.

Task SMSE averages D mean squared errors divided by the corresponding A-target variances across horizons. Nonlinear repetition loss averages the three nonlinear laws and both noise levels; the linear panel is separate. The five nonlinear margin contrasts are $d_r = 0.97L^{\mathrm{NL}}_{r,m} - L^{\mathrm{NL}}_{r,K}$; the linear contrast is $d_r = 1.05L^{\mathrm{lin}}_{r,\mathrm{ridge}} - L^{\mathrm{lin}}_{r,K}$. Each criterion requires $\bar{d} - t_{23,1-0.05/6}s_d/\sqrt{24} > 0$. This is an approximate one-sided Bonferroni paired-t

Table 3: The preceding dimension study's unchanged six-claim family. L_K, L_m are mean SMSEs. Gain is $100(1 - L_K/L_m)$. The first five rows test a 3% nonlinear improvement and the last a 5% linear noninferiority tolerance. The simultaneous lower endpoint concerns the absolute margin contrast; a positive endpoint establishes the criterion. The complete ten-procedure results remain in the appendix.

Comparator	L_K	L_m	Gain (%)	$\bar{d}$	Lower bound	Established
Common features	0.02891	0.03178	9.01	0.001910	0.001529	Yes
Bounded RBF departure	0.02891	0.02912	0.72	-0.000663	-0.000800	No
Raw RBF	0.02891	0.03143	8.01	0.001574	0.001156	Yes
Boosted trees	0.02891	0.02962	2.39	-0.000181	-0.000662	No
Selected non-KAHM	0.02891	0.02778	-4.08	-0.001967	-0.002218	No
Raw ridge (linear NI)	0.90342	0.90166	-0.20	0.043324	0.041880	Yes

calculation across 24 independent repetition aggregates, not across dependent origins. The declared joint criterion requires all six comparisons and the separate numerical screens to pass.

KAHM's nonlinear loss is 0.028912. Its gains over common features and raw RBF, 9.01% and 8.01%, establish their 3% margins. Gains over independently selected RBF departures (0.72%) and boosting (2.39%) do not; loss is 4.08% higher than the selected non-KAHM procedure. Linear noninferiority passes: three of six criteria are established, so the joint performance criterion fails. An unestablished margin does not imply equality.

Dimension selection lowers loss relative to fixed cap two by 3.26%, whereas native cap twenty raises it by 8.39%. On the KAHM-selected bank, the total loss change -0.000973 comprises a common-feature improvement -0.002150 and a weakening of the incremental departure benefit by $+0.001177$. This descriptive fitted-loss decomposition is not a causal or orthogonal variance decomposition. Matched-bank KAHM improves over RBF by 0.68%, without establishing the primary 3% margin. All 192 tasks complete, the feature and correction bounds have no violations, and the maximum taskwise KAHM-to-ridge maximum-error ratio is 1.4673, below five. These checks establish computational coverage, not forecasting superiority.

7.2 Preprocessing and residual-target follow-ups

Both follow-ups use the same four laws, width-12 histories, horizons $1, 4, 12$, and 192 tasks per cohort. Preprocessing has 24 fresh repetition units. The residual-target results are a same-cohort computational reproduction of the original 24 units, with 0 additional independent repetitions. Each retains a separate four-claim family, one-sided Bonferroni paired-t bounds with 23 degrees of freedom, 3% nonlinear superiority margins, and 5% linear noninferiority.

Preprocessing. B jointly selects transformation and native cap. Candidate transforms are standardization, unclipped min–max scaling, and tanh of standardized histories; the five common/departure pairs include both min–max/tanh hybrids. The selected KAHM procedure has 0.28% higher nonlinear loss than independently selected RBF and 4.02% higher loss than the non-KAHM policy. Tanh/tanh has 6.49% higher loss than standard/standard. Only linear noninferiority passes, despite observed excess loss 0.18%: 1 of 4 criteria. A separate secondary design fits a common forecast on 32 episodes and corrections on 32 disjoint episodes. Across its five scaler pairs, KAHM reduces loss by 10.78–14.93% versus the identical common-plus-bias baseline and 2.37–4.14% versus independently selected RBF. These descriptive gains do not reverse the primary result.

Residual targets. The next design retains a B-selected common forecast fitted on all 64 A episodes and a shared B residual-mean bias β_h:

$$\widehat{f}_{m,h}(x) = f_{\text{base},h}(x) + \eta_{m,h}\delta_{m,h}(x), \qquad f_{\text{base},h} = f_{\text{com},h} + \beta_h, \quad \eta_{m,h} \in \{0, \tfrac{1}{4}, \tfrac{1}{2}, 1\}. \tag{16}$$

The centered bounded departure slope $\delta_{m,h}$ adds no residual-mean intercept. Four episode-wise folds produce out-of-fold (OOF) targets from 48-episode common predictors; the in-sample alternative uses the 64-episode deployed common predictor. Both use full-A departure banks, with independently B-selected scalers, caps

and penalties and C-selected multipliers. The comparison changes residual generation and resulting selection, not just one target vector at fixed settings.

OOF KAHM improves the shared baseline by 9.20%, establishing its 3% margin. The 2.87% gain over independently selected OOF RBF does not establish that margin; loss is 5.12% higher than the expanded non-KAHM policy. Linear noninferiority passes with observed excess 1.13%: 2 of 4 criteria. The secondary in-sample correction has SMSE 0.027559, improving the shared baseline by 12.43% and independent in-sample RBF by 3.96%, yet remaining 1.39% above the policy in a post hoc mean comparison. OOF KAHM is 3.69% worse than in-sample KAHM, which is lower in all 24 nonlinear repetition aggregates. Different fold-training sizes and fitted settings limit causal attribution.

Original completed residual-study records were unavailable; the reproduction uses the prescribed streams and unchanged scientific code with prior outcome knowledge. Its audit verifies the newly retained execution, including 8,064 forecast arrays, without authenticating the missing original records or creating a fresh confirmatory cohort. Secondary comparisons retain their descriptive status irrespective of their effect sizes.

7.3 Policy augmentation and a separate natural panel

A subsequent frozen policy study asks whether making the in-sample KAHM correction available improves selection. It uses 24 fresh repetitions and 192 tasks under the same laws and A/B/C/D roles; earlier outcomes informed its design. P_0 selects one family from the ten-family non-KAHM menu on C. P_1 adds the centered KAHM correction in Equation (16); P_0^+ instead adds a centered raw-history Nyström RBF correction. Both additions retain the same common forecast and shared bias and receive fifteen feature configurations, five ridge penalties, and one added C-selection slot. Equal search counts do not match width, statistical capacity, or runtime. C calibrates each horizon and chooses a complete family; ties retain the existing menu. A lower selected C loss need not imply lower D loss.

Nonlinear mean SMSEs are 0.02726839 for P_0, 0.02704583 for P_1, and 0.02722502 for P_0^+. The gains 0.82% and 0.66% fall below the prescribed 3% margins. For $d_r = L_{P_1,r} - aL_{Q,r}$, where $a = 0.97$ for nonlinear superiority and $a = 1.05$ for linear noninferiority, the separate four-claim family requires

$$\bar{d} + t_{23,1-0.05/4}\, s_d/\sqrt{24} < 0. \tag{17}$$

Both linear noninferiority criteria and neither nonlinear superiority criterion are established (Table 4). This approximate repetition-level paired-t inference does not exclude a smaller benefit.

KAHM is selected in 44 tasks: 43 nonlinear and one linear. Among the nonlinear selections, thirty improve D loss and thirteen worsen it; these conditional counts are descriptive. Standalone KAHM improves the identical common-plus-bias baseline by 12.50% and independent local RBF by 3.83%, but has 1.32% higher mean nonlinear loss than P_0. Useful residual correction therefore does not establish the declared policy advantage.

Natural-series evaluation. A separate panel uses 24 Weekly and 24 Hourly M4 training series, chosen by fixed ID hashes and length eligibility while excluding 72 earlier pilot IDs (M4 Competition, 2026). No official Test data are used. Chronological A/B/C/D blocks have twelve intervening origins; D has 32 rolling origins at horizons $1, 4, 12$. Every target in one block is observed before the next begins. Models remain fixed; later histories may include earlier outcomes already observed at that origin. Seasonal naive and drift replace the episode-OOF RBF family, giving eleven base families and twelve per enlarged menu.

Overall SMSE is 0.127612 for P_0, 0.130512 for P_1, and 0.134159 for P_0^+. Augmentation is 2.27% worse than the base policy, despite being 2.72% better than the alternative enlargement. Only 3 of 48 decisions change: one improves and two worsen; the other 45 forecasts equal P_0. Losses average horizons within series, series within frequency, and frequencies equally; scales use A alone. Anonymous series may share sources, so these results have no independence-based confidence interval. They are descriptive benchmark-development evidence, not an official M4 ranking or a new population-level superiority claim.

The audits cover 9,600 synthetic and 2,016 natural forecast arrays. Synthetic source publication preceded terminal generation; the natural protocol was frozen locally before parsing values and published after evaluation. Neither is public preregistration. Complete grids, primary bounds, selected cases, frequency

Table 4: The preceding policy study's four original criteria, preserved separately from the representation ablation. Gain is $100(1 - \overline{L}_{P_1}/\overline{L}_Q)$; the adjusted upper endpoint concerns the absolute margin contrast in Equation (17). A negative endpoint establishes the criterion.

Panel	Comparator	Margin	Gain (%)	Upper bound	Established
Nonlinear	P_0	3%	+0.8162	+0.000711	No
Nonlinear	P_0^+	3%	+0.6582	+0.000798	No
Linear	P_0	5%	-0.0016	-0.044648	Yes
Linear	P_0^+	5%	+0.0016	-0.044677	Yes

losses, and audit limits remain in the *Supplementary Studies*. Any selector redesigned from these terminal outcomes requires newly reserved evaluation.

7.4 Earlier evidence and limits of transfer

The older investigations use their own geometric encodings, rank policies, targets and inferential families; they are not interchangeable instances of the present soft-state map. The *Supplementary Studies* retains their full methods, tables, proofs, and failure accounting.

Controlled target interventions show why reconstruction fidelity alone can lose predictive position: an affine reconstruction can be nearly exact while a discrepancy-based code discards the relevant coordinate. Local position restores prediction in that example. Separate finite-decoder comparisons favor KAHM on Van der Pol but favor controls on nonlinear autoregression or Duffing. Thus their positive findings depend on the law and decoder.

On four reserved natural source families, raw-history augmentation with KAHM has 4.26% higher source-balanced loss than raw RBF and costs 2.30 times as much; ridge is more accurate than both. Post hoc scaling changes preserve the principal aggregate ordering. Additive corrections yield some synthetic gains, but eight failed geometric fits and four unavailable policy outputs prevent their planned complete archive comparison. A post hoc successful-subset gain does not repair that missing aggregate.

The learned-metric donor-residual study completes all 48 tasks and improves on ridge, but its metric-versus-identity and donor-versus-in-sample intervals include zero, and no adjusted comparison against an equally trained geometric control favors KAHM. Direct reconstruction training has two favorable, six unfavorable and ten unresolved primary contrasts; neither favorable result is against an equally trained local control. Three infeasible rideshare series leave its planned natural aggregate undefined. Both online capacity settings fail their accuracy criteria, and severe extrapolation persists after a solver correction. The present bounded fixed-geometry decoder is a different procedure; its bound does not retrospectively validate the adaptive method.

These outcomes constrain transfer beyond the specified constructions. Their observations are not pooled with either the later forecasting studies or the new representation ablation.

8 Discussion and conclusion

The four-arm comparison gives the soft-state viewpoint a specific empirical question: which geometric observables help a fixed family of readouts predict an external target? P versus G isolates access supplied by local position; K versus P asks whether KAHM departures add useful functions; K versus R asks whether that increment exceeds a matched positive-kernel reconstruction. These are different claims. Success for position does not establish a KAHM-specific advantage, and an advantage over memberships alone does not establish sufficient state representation.

The complete primary family establishes three of four criteria. Position reduces nonlinear loss by 69.20% relative to memberships, and KAHM departures reduce P's loss by 10.46%; both exceed their 3% margins under the simultaneous bounds. The K/R gain is only 0.03%, so the KAHM-specific margin is unestablished. K's

linear loss is 1.11% higher than P's, while meeting the specified 5% noninferiority criterion. Consequently, the strongest supported claim concerns useful within-region position and conditional reconstruction enrichment. It does not identify inverse-Gram reconstruction as generally preferable to the matched positive-kernel construction. Nor does a large P/G gain show that memberships have no predictive content: it shows their limitations for this affine readout and allocation.

The primary bank is selected for P, so the comparison concerns enrichment of a validation-selected common representation. It does not optimize the bank independently for G, K, or R. The secondary independent-cap analysis asks a different question about complete procedures. A discrepancy between contexts can identify sensitivity to geometry selection; it cannot replace a failed primary criterion. Likewise, failure to establish a margin is not evidence of equivalence. The block transformations preserve shared coordinates, but fitted access still depends on radial scaling, regularization, sample size, and the affine readout. The comparison does not estimate all information retained by an abstraction.

The secondary results reinforce that distinction. Independently selected K and R attain nonlinear mean losses 0.0267546 and 0.0267507 the K/R point difference changes sign without establishing either equivalence or a new primary claim. K improves P on the oscillators but worsens it on nonlinear autoregression, and its aggregate advantage is concentrated at the longest tested horizon. The selected enrichments have more coordinates than the original history, so predictive utility here is not evidence of compression. The recorded costs cover the full candidate search with shared computations; they do not establish equal computation between K and R or an isolated cost advantage for either.

The raw-span results explain why a change in reconstruction rule need not create a new predictive function. Matched KAHM and RBF augmented spans coincide under the stated symmetry conditions, while irregular reference configurations can separate them. Equality concerns jointly variable coefficients, including common coordinates. It does not equate departures added to a frozen forecast, nor does it survive arbitrary nonlinear transformations as a general rule. Conversely, a distinct raw function need not align with the target or be estimable. Equal nominal width and a common ridge architecture therefore provide useful controls without proving equal effective capacity, conditioning, or computational cost.

Local dimension acts on several parts of this construction at once: position coordinates, projected kernels and reconstruction departures. The historical dimension decomposition finds stronger common prediction and a smaller incremental departure benefit after selection. Native full ambient rank is admissible, with caps twelve and twenty coinciding at history width twelve; neither fact guarantees accuracy. The preceding preprocessing and residual-target results likewise discourage universal rules about tanh scaling or cross-fitting. Cross-fitted common predictors use fewer episodes and may choose different settings, so their comparison with an in-sample correction does not identify the effect of residual fitting alone.

Representation-only readouts also delimit what the earlier residual gains establish. A retained raw-history predictor can supply information absent from its correction features. Within that architecture, a correction helps when its alignment with the retained residual outweighs its squared magnitude. Generic ridge arguments bound the correction, not the common forecast or total error. Radial bounding does not remove sensitivity at signed-normalization fallback boundaries. These mathematical controls apply to matched alternatives and establish neither forecasting superiority nor rollout stability.

The completed policy study illustrates a further distinction. Its standalone KAHM correction improves the common-plus-bias baseline by 12.50% and local RBF by 3.83%, yet remains 1.32% worse than the base policy. Admitting it produces gains of 0.82% and 0.66% against the base and enlarged-RBF policies, establishing neither superiority margin. Equal search counts control added selection opportunities without equating cost or capacity. The natural panel changes only three of forty-eight decisions, improving one and worsening two, with a 2.27% aggregate penalty against the base policy. Its potentially dependent sources and limited frequency coverage support descriptive conclusions, not an official M4 ranking or a population-level superiority claim.

Coordinate dynamics remain a separate requirement. Proposition 4 applies the standard observable-space criterion for exact deterministic closure to the raw-span comparison. It does not exhibit an invariant representation for the experimental processes. Under stochastic transitions, a linear conditional-mean

evolution can coexist with irreducible realized-successor variation; iteration requires appropriate Markov and time-homogeneity assumptions. The companion certificates (Kumar & Kargaran, 2026a) specify a fixed simplex map, evaluation law and norm-bounded prediction class. They do not automatically apply to signed enriched features or potentially unbounded observation residuals.

A propagation investigation would therefore need to evaluate both direct and propagated readouts against the same observed outputs, alongside held-out successor errors and sensitivity to the horizon. Predicting each candidate's own coordinates changes the target with the representation. Nearly constant coordinates may be easy to propagate while discarding useful distinctions; low coordinate error or an inconclusive lower certificate alone cannot establish utility. Such evidence is necessary for a claim about useful autonomous soft-state dynamics, rather than for the narrower direct-readout comparisons considered here.

Further selector design should depend on available incremental opportunity. Saved forecasts can quantify that opportunity and losses from missed or harmful selections exploratorily. Limited opportunity would direct attention to the observables or target alignment; substantial opportunity would motivate a better selector. Any design informed by terminal outcomes requires newly reserved evaluation. Fresh repetitions of familiar laws, historical adaptive failures, and limited modern forecasting benchmarks constrain the external scope of either conclusion.

Geometry-induced soft-state abstractions are therefore a useful organizing framework when membership, position and reconstruction are evaluated as distinct observables. The present experiment supports the predictive value of position and a conditional KAHM departure increment, while leaving a general KAHM-specific advantage over matched RBF unestablished. The raw-span theory explains possible functional distinctions without turning them into a guarantee of utility. Useful autonomous dynamics and practical policy gains remain separate empirical questions. The evidence supports a representation research program with explicit external targets and matched controls, rather than a general forecasting-superiority claim.

Reproducibility and research transparency

The four-arm implementation and analysis specification are maintained under `extensions/soft_state_ablation_20261102`. The protocol requires source publication before production preparation, a sealed A/B fitted-state roster before D generation, and sealed terminal forecasts before scoring. The fixed budget, primary family and secondary contexts are specified before production outcomes. Artificial checks cover shared blocks, full and zero rank, cap aliases, ridge paths, portable state, numerical recovery, and staged execution. A separate artificial audit reconstructs eight miniature tasks, and a deterministic 32-repetition loss fixture checks the statistical design. These checks test implementation behavior, not forecasting performance.

Scientific source commit `2b7d19b5` was published and its twenty-five files verified at 09:20:24 UTC on 2 November 2026. The source-freeze SHA256 begins `2745d10525a8fbc1`; the complete digest is retained in `PUBLICATION.json`, and the source list in `freeze.json`. Preparation finished at 09:20:57, all fits were sealed at 09:28:00, D preparation finished at 09:28:33, and predictions were sealed at 09:34:57. Target release followed at 09:35:11 and analysis at 09:35:18. These are recorded procedural controls, not independent preregistration, oversight or cryptographic target isolation.

The independent production audit passed 2563814 checks covering all 256 tasks. It independently refitted all 25,600 B penalty-path forecast arrays using Cholesky normal equations, and replayed 2,048 selected C and 2,048 selected D arrays from the portable fitted geometry. Maximum absolute prediction differences were 1.20×10^{-11}, 2.43×10^{-12} and 5.59×10^{-12} respectively. It reconstructed histories and targets from saved base episodes and reserved streams, checked native dimensions and exact shared-coordinate identities, and independently recomputed the primary and descriptive statistics. It did not independently refit partition k-means or reintegrate the physical laws. All planned tasks completed, with zero solver recoveries and zero query-denominator fallback calls. One singleton local model used the declared rank-zero wrapper at all five caps.

Compact results and the audit summary are published with `analysis/` and `verification/`; lossless compressed files retain the largest tables and full audit receipt. Their restoration script verifies hashes before writing. An indexed numerical archive retains full fitted states, forecasts, data and candidate paths, while

the public source can regenerate them. The paper includes the generated display assets and can compile without numerical archives or refitting. The completed result publication is `65ecf679`; it is separate from the frozen scientific source and the subsequent manuscript edition.

Historical evidence retains separate records and inferential units. The dimension-study audit checks 192 tasks, 18,816 local bases and 27 analysis tables under `extensions/dimension_forecasting`. The preprocessing study and residual reproduction are under `extensions/preprocessing_dimension` and `extensions/crossfit_residual_reproduction_20261101`. The reproduction restores a prescribed computation with prior outcome knowledge; it neither recovers missing original run records nor adds a fresh confirmatory cohort. Its zero-rank verification correction and numerical recoveries have separate provenance.

The policy records remain under `extensions/policy_augmentation_20261101` and `extensions/policy_augmentation_natural_20261101`. Scientific source `b1915813` and result publication `b5941040` identify that completed study. Its simulation audit checks 192 tasks and 9,600 terminal arrays; the natural audit checks chronology, 2,016 arrays and 144 policy choices. Those audits reuse inherited feature maps and do not independently refit every unselected candidate. Their detailed coverage and limits remain in the Supplementary Studies.

Displayed empirical values are generated from the corresponding study tables with input and output hashes. Full principal proofs and the earlier dimension protocol remain in the main appendices; the Supplementary Studies preserves later protocols and older investigations. Source and manuscript editions have separate identities. A document build, checksum verification, forecast replay and independent refitting are distinct checks; none constitutes external scientific replication.

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A Reconstruction algebra and bounded residual decoding

This appendix analyzes fixed reference maps and their use by a specified decoder. It establishes normalized-kernel identities, conditions for equality or inequality of feature spans, and a bound on fitted residual corrections. None of these statements assumes a particular time-series distribution or guarantees a forecasting improvement. Unless another subscript is displayed, vector norms are Euclidean norms. The numerical examples can be evaluated with `extensions/reconstruction_mechanism/algebra.py`; their reference banks are constructed independently of forecasting outcomes.

A.1 Normalized kernel identities

Let $q_1, \dots, q_M \in \mathbb{R}^d$ be reference histories, and let $Q \in \mathbb{R}^{M\times d}$ contain their transposes as rows. Let $P \in \mathbb{R}^{d\times r}$ have orthonormal columns, with $1 \le r \le d$, and let $W \in \mathbb{R}^{r\times r}$ be a symmetric positive-definite precision matrix for the projected coordinates. For a column-vector query $z \in \mathbb{R}^d$, define the Gaussian kernel vector $k(z) \in \mathbb{R}^M$ by

$$k_i(z) = \exp\left[-\frac{(P^\top(z-q_i))^\top W(P^\top(z-q_i))}{2r}\right], \qquad i = 1, \dots, M.$$

Let $K \in \mathbb{R}^{M\times M}$ be the reference Gram matrix, with $K_{ij} = k_i(q_j)$. For $\lambda > 0$, set $C = (K + \lambda I_M)^{-1}$, where I_M is the $M \times M$ identity matrix, and let $\mathbf{1}_M$ be the length-M vector of ones. The matrix K is symmetric positive semidefinite, so C exists and is symmetric positive definite. The KAHM and matched positive-kernel reconstructions are

$$A_K(z) = \frac{Q^\top Ck(z)}{\mathbf{1}_M^\top Ck(z)}, \qquad A_R(z) = \frac{Q^\top k(z)}{\mathbf{1}_M^\top k(z)}. \tag{18}$$

The first expression is defined wherever its signed denominator is nonzero. The second denominator is positive for every finite query. The two maps share the references and projected Gaussian kernel. Multiplying $k(z)$ by a common positive scalar cancels in either ratio, which justifies subtracting the largest log-kernel value before query evaluation.

Target-transfer identity. For a query satisfying $s(z) = \mathbf{1}_M^\top Ck(z) \neq 0$, define the normalized weight vector $\omega(z) = Ck(z)/s(z)$. Its entries sum to one. For any reference target vector $y \in \mathbb{R}^M$,

$$\omega(z)^\top y = \frac{k(z)^\top (K + \lambda I_M)^{-1} y}{k(z)^\top (K + \lambda I_M)^{-1}\mathbf{1}_M}.$$

Thus target transfer is the ratio of two dual kernel-ridge predictions with regularization matrix $K + \lambda I_M$ and no additional intercept: one fitted to y, and one fitted to the constant target $\mathbf{1}_M$. For an affine scalar function $f(q) = q^\top b + c$, with $b \in \mathbb{R}^d$ and $c \in \mathbb{R}$, let $f(Q) = Qb + c\mathbf{1}_M$ be its reference values. Then

$$\omega(z)^\top f(Q) - f(z) = \{A_K(z) - z\}^\top b.$$

Proof of Proposition 1. The definition gives $\mathbf{1}_M^\top \omega(z) = s(z)/s(z) = 1$. Because $C = C^\top$, $\omega(z)^\top y = k(z)^\top Cy/s(z)$ and $s(z) = k(z)^\top C\mathbf{1}_M$, proving the ratio identity. For the affine function,

$$\omega(z)^\top f(Q) = \omega(z)^\top Qb + c\omega(z)^\top \mathbf{1}_M = A_K(z)^\top b + c.$$

Subtracting $z^\top b + c$ proves the final identity. □

This identifies direct target transfer as a normalized kernel-ridge construction. The ratio generally differs from the unnormalized kernel-ridge predictor because its denominator depends on the query. Similarly, $A_K(z)^\top b$ is normalized kernel ridge applied to the restricted reference target Qb. For any reconstruction A, the identity $A(z) + \{z - A(z)\} = z$ also implies a linear dependence among the raw input, reconstruction, and residual when all three are included as unrestricted linear features. These identities constrain claims about the resulting function class while allowing useful parameterizations for a restricted or regularized decoder.

A.2 When reconstruction departures have the same span

Let $\mu \in \mathbb{R}^d$ be the reference center and define the affine reconstruction $A_A(z) = \mu + PP^\top(z - \mu)$. Fix a common scalar locality gate $g(z)$, and define column-vector departures

$$d_K(z) = g(z)\{A_K(z) - A_A(z)\}, \qquad d_R(z) = g(z)\{A_R(z) - A_A(z)\}.$$

A feature span means the set of scalar functions obtained by finite linear combinations of its coordinate functions. Let $\mathcal{S}_0$ be the span of the gate $g(z)$ and the coordinates of $g(z)P^\top(z - \mu)$. It contains every coordinate of $g(z)A_A(z)$.

Equal augmented feature spans. Suppose $K\mathbf{1}_M = \kappa\mathbf{1}_M$ for a scalar κ and C leaves the augmented reference-coordinate space $\mathcal{V} = \mathrm{span}\{\mathbf{1}_M, Q_{:1}, \ldots, Q_{:d}\} \subseteq \mathbb{R}^M$ invariant. Here $Q_{:j}$ denotes column j of Q. Then there exist $B \in \mathbb{R}^{d\times d}$ and $c \in \mathbb{R}^d$ such that

$$C\mathbf{1}_M = a\mathbf{1}_M, \qquad CQ = QB + \mathbf{1}_M c^\top, \qquad a = (\kappa + \lambda)^{-1} > 0.$$

For these coefficients,

$$A_K = \frac{B^\top A_R + c}{a}, \qquad d_K = \frac{B^\top}{a} d_R + \left(\frac{B^\top}{a} - I_d\right) gA_A + \frac{gc}{a}.$$

The span obtained by adjoining the coordinates of d_K to $\mathcal{S}_0$ is the same as that obtained by adjoining the coordinates of d_R.

Proof of Proposition 2. The constant-row-sum assumption implies $(K+\lambda I_M)\mathbf{1}_M = (\kappa+\lambda)\mathbf{1}_M$, so $C\mathbf{1}_M = a\mathbf{1}_M$. Since K is positive semidefinite, $\kappa \geq 0$ and hence $a > 0$. Invariance of $\mathcal{V}$ means each column of CQ can be expressed using Q and $\mathbf{1}_M$, which gives B and c. Using symmetry of C in Equation (18),

$$A_K(z) = \frac{(CQ)^\top k(z)}{(C\mathbf{1}_M)^\top k(z)} = \frac{B^\top Q^\top k(z) + c\mathbf{1}_M^\top k(z)}{a\mathbf{1}_M^\top k(z)} = \frac{B^\top A_R(z) + c}{a}.$$

Subtract A_A and multiply by g to obtain the displayed departure identity. Since $gA_A = g\mu + P\{gP^\top(z - \mu)\}$, all its coordinates belong to $\mathcal{S}_0$. Thus the KAHM-augmented span is contained in the RBF-augmented span.

For the reverse inclusion, the restriction of the invertible matrix C to the invariant finite-dimensional space $\mathcal{V}$ is injective and therefore surjective onto $\mathcal{V}$. Its images $C\mathbf{1}_M = a\mathbf{1}_M$ and the columns of CQ consequently span $\mathcal{V}$. Hence there exist $D \in \mathbb{R}^{d\times d}$ and $e \in \mathbb{R}^d$ such that $Q = (CQ)D + \mathbf{1}_M e^\top$. Multiplying its transpose by $k(z)/(\mathbf{1}_M^\top k(z))$ gives $A_R(z) = aD^\top A_K(z) + e$. After multiplication by g and subtraction of gA_A, this expresses each RBF departure coordinate using KAHM departure coordinates and $\mathcal{S}_0$. The spans therefore agree, including when the original reference-coordinate columns are redundant. □

For example, take scalar references $q = (-1, 1)^\top$, $P = 1$, $g = 1$, and $A_A(z) = z$. If the off-diagonal Gram entry is ρ, then $K\mathbf{1}_2 = (1 + \rho)\mathbf{1}_2$ and $Kq = (1 - \rho)q$. Therefore

$$A_K = c_\rho A_R, \qquad d_K = c_\rho d_R + (c_\rho - 1)z, \qquad c_\rho = \frac{1 + \rho + \lambda}{1 - \rho + \lambda}. \tag{19}$$

The coefficient c_ρ is positive, so $\mathrm{span}(1, z, d_K) = \mathrm{span}(1, z, d_R)$. With the sample-variance precision $1/2$, $\rho = e^{-1}$. The regularization iteration used by the implementation yields $\lambda = 2.6519665429866075$ and

$c_\rho = 1.224037566462905$. The maximum numerical identity error on the constructed query grid is below 6×10^{-16}.

These are equalities between untransformed feature spans. An isotropic ridge penalty is not invariant under a general nonorthogonal change of coordinates. Applying separate nonlinear radial transformations can also remove the displayed linear relation. A forecasting comparison therefore evaluates the complete reconstruction, feature transformation, and fitting procedure, even when two untransformed feature sets have the same span.

A.3 Freezing the common predictor

Consider the two-reference example in Equation (19) on $\mathbb{R}$. For a Gaussian kernel of positive precision, $0 < \rho < 1$, so $c_\rho > 1$. Writing $t > 0$ for the scalar precision gives $A_R(z) = \tanh(tz)$, which is bounded and nonconstant. If $az + bd_R(z) = 0$, then $(a-b)z + bA_R(z) = 0$. Boundedness first gives $a = b$, and nonconstancy then gives $b = 0$. Thus z and d_R are linearly independent. Were $d_K = ud_R$ for a scalar u, Equation (19) would give $(c_\rho - 1)z + (c_\rho - u)d_R = 0$, contradicting that independence. The nonzero one-dimensional spans of d_K and d_R are therefore distinct and neither contains the other. Translating both by any fixed f_{base} proves the claims about Equation (8). Permitting the coefficient of z to vary restores the equal augmented spans already proved above.

For a fixed fitting sample, define $H_A v(z) = v(z) - \bar{v}_A$. Linearity gives

$$H_A d_K = c_\rho H_A d_R + (c_\rho - 1) H_A z.$$

The same non-equality argument applies whenever $H_A z$ and $H_A d_R$ are linearly independent on the query domain under consideration. On the whole real line this follows from independence of $1, z, d_R$: boundedness and nonconstancy of A_R prove it by the preceding argument with an added constant. Independence need not hold on an arbitrary finite query set. Separate nonlinear radial maps need not preserve the displayed identity. These are function-class statements; they give no risk ordering for fitted or calibrated forecasts.

A.4 An explicit distinct reconstruction basis

Irregular scalar references. Let $q_1 < q_2 < q_3$ with $q_1 + q_3 \neq 2q_2$, and define $k_i(z) = \exp\{-t(z - q_i)^2/2\}$ for precision $t > 0$. Let $K_{ij} = \exp\{-t(q_i - q_j)^2/2\}$, $C = (K + \lambda I_3)^{-1}$ for $\lambda > 0$, and suppose every entry of $v = C\mathbf{1}_3$ is positive. For the reconstructions in Equation (18), with $Q = q = (q_1, q_2, q_3)^\top$, gate $g = 1$, and $A_A(z) = z$,

$$A_K \notin \operatorname{span}(1, z, A_R), \qquad d_K \notin \operatorname{span}(1, z, d_R).$$

These are statements about scalar functions on the whole real line.

Proof of Proposition 3. Set $u = Cq$. Since C is symmetric, $A_K(z) = (k(z)^\top u)/(k(z)^\top v)$, and positivity of v makes its denominator positive for every z. The six sums $q_i + q_j$ with $i \leq j$ are distinct: strict ordering of the three references orders all the sums except possibly $q_1 + q_3$ and $2q_2$, and their equality is excluded by assumption. Products of Gaussian translates satisfy

$$k_i(z)k_j(z) = e^{-tz^2} e^{-t(q_i^2 + q_j^2)/2} e^{t(q_i + q_j)z}.$$

After removal of the common positive factor e^{-tz^2}, these products are nonzero multiples of exponentials with six distinct real exponents. Such exponentials are linearly independent. Indeed, differentiating a vanishing linear combination zero through fifth order at $z = 0$ gives a Vandermonde system whose determinant is the product of the nonzero pairwise exponent differences.

Suppose first that $A_K = cA_R + d$ for scalars c, d. Define $w = u - dv$. Multiplication by the two positive denominators gives

$$\left(\sum_i k_i w_i\right)\left(\sum_j k_j\right) = c\left(\sum_i k_i q_i\right)\left(\sum_j k_j v_j\right).$$

Independence of the six products implies equality of their coefficients. The diagonal coefficients give $w_i = cq_iv_i$, and the off-diagonal coefficients give $w_i + w_j = c(q_iv_j + q_jv_i)$ for $i < j$. If $c = 0$, all w_i vanish, so $Cq = dC\mathbf{1}_3$. Invertibility of C would imply $q = d\mathbf{1}_3$, contrary to the distinct references. If $c \neq 0$, substituting the diagonal equations into the off-diagonal ones yields $(q_i - q_j)(v_i - v_j) = 0$. Thus $v = a\mathbf{1}_3$ for some $a > 0$, and $(K + \lambda I_3)v = \mathbf{1}_3$ implies that K has constant row sums. This is impossible: its middle-row sum exceeds the first-row sum by $K_{23} - K_{13} > 0$ and the last-row sum by $K_{12} - K_{13} > 0$, since the Gaussian decreases strictly with positive distance. This rules out every affine relation $A_K = cA_R + d$.

Both reconstructions are bounded functions of z. The RBF reconstruction is a convex combination of q_1, q_2, q_3. Also,

$$A_K(z) = \frac{\sum_i k_i(z)v_i\,(u_i/v_i)}{\sum_i k_i(z)v_i}$$

is a convex combination of the three finite numbers u_i/v_i. If $A_K(z) = cA_R(z) + d + bz$ held on the real line, boundedness of both reconstructions would force $b = 0$. The affine relation already ruled out would then follow. This proves the first span statement. Finally, $d_K = A_K - z$ and $d_R = A_R - z$ imply $\operatorname{span}(1, z, d_R) = \operatorname{span}(1, z, A_R)$. Membership of d_K in this span would imply membership of A_K, a contradiction. □

A concrete example uses $q = (-1, 0, 2)^\top$, sample-variance precision $t = 3/7$, and $\lambda = 4.498274743339644$ from the specified regularization iteration. Positivity of $C\mathbf{1}_3$ can be verified without examining a query grid. Write $K = I_3 + E$ and $a = 1 + \lambda$. For a matrix, let $\|\cdot\|_\infty$ denote the maximum absolute row sum; for a vector, it denotes the largest absolute entry. If $a > 2\|E\|_\infty$, then the Neumann series for $aC = (I_3 + E/a)^{-1}$ gives

$$\|aC\mathbf{1}_3 - \mathbf{1}_3\|_\infty \le \sum_{j=1}^{\infty}(\|E\|_\infty/a)^j = \frac{\|E\|_\infty/a}{1 - \|E\|_\infty/a} < 1.$$

Every entry of $aC\mathbf{1}_3$ is then strictly positive. In the example, $\|E\|_\infty = e^{-3/14} + e^{-6/7} = 1.2314905926823394$, so the condition holds. On 81 constructed queries in $[-3, 4]$, the maximum absolute residual after least-squares projection of the sampled d_K values onto the sampled $(1, z, d_R)$ columns is 0.05355670997078987. This numerical illustration supplements the analytic existence result. It does not establish distinctness for every higher-dimensional bank used in the forecasting experiments.

A.5 A bounded correction with a free intercept

For a feature block $v \in \mathbb{R}^{p_b}$ of positive width p_b, define

$$T_{R_b}(v) = \frac{v}{\sqrt{1 + \|v\|_2^2/R_b^2}}, \qquad R_b = \sqrt{p_b}. \tag{20}$$

Its norm is at most R_b. To see its sensitivity to its own input, set $a = 1 + \|v\|_2^2/R_b^2$. Its Jacobian is $a^{-1/2}I - a^{-3/2}vv^\top/R_b^2$. For $v \neq 0$, the eigenvalue in the direction of v is $a^{-3/2}$, and each orthogonal direction has eigenvalue $a^{-1/2}$; at zero, the Jacobian is the identity. All its singular values are therefore at most one. Integrating the derivative along the segment between any two inputs proves $\|T_{R_b}(v) - T_{R_b}(v')\|_2 \le \|v - v'\|_2$. Thus the radial transformation is nonexpansive.

Concatenating separately transformed blocks gives a feature vector $\psi(z) \in \mathbb{R}^p$, where $p = \sum_b p_b$, satisfying $\|\psi(z)\|_2 \le R = (\sum_b R_b^2)^{1/2}$. This norm bound holds wherever the preceding feature construction is defined. Nonexpansiveness of the radial transformation does not by itself give a Lipschitz constant for the full history-to-feature map, which includes the signed reconstruction denominator and its fallback rule.

Centered ridge correction. For $n \ge 1$, let $e_i \in \mathbb{R}$ be residual targets and $\psi_i \in \mathbb{R}^p$ be training feature vectors. Fit $b \in \mathbb{R}$ and $w \in \mathbb{R}^p$ by minimizing

$$\frac{1}{n}\sum_{i=1}^{n}(e_i - b - \psi_i^\top w)^2 + \alpha\|w\|_2^2, \qquad \alpha > 0,$$

with no penalty on b. Define the training means $\bar{e} = n^{-1}\sum_i e_i$ and $\bar{\psi} = n^{-1}\sum_i \psi_i$, the residual vector $e = (e_1, \ldots, e_n)^\top$, its centered version $e_c = e - \bar{e}\mathbf{1}_n$, and its centered root mean square $\sigma_e = \|e_c\|_2/\sqrt{n}$. Suppose query features satisfy $\|\psi(z)\|_2 \le R$. Then, for any scalar γ,

$$\|w\|_2 \le \frac{\sigma_e}{2\sqrt{\alpha}}, \qquad |\gamma q(z)| \le |\gamma| \left[|\bar{e}| + (R + \|\bar{\psi}\|_2)\frac{\sigma_e}{2\sqrt{\alpha}} \right], \quad q(z) = \bar{e} + (\psi(z) - \bar{\psi})^\top w. \tag{21}$$

Proof of Proposition 5. For fixed w, differentiating with respect to the intercept gives $b = \bar{e} - \bar{\psi}^\top w$. Let $\Psi_c \in \mathbb{R}^{n\times p}$ have row i equal to $(\psi_i - \bar{\psi})^\top$. Substituting the fitted intercept reduces the objective to $n^{-1}\|e_c - \Psi_c w\|_2^2 + \alpha\|w\|_2^2$. Its unique minimizer is

$$w = (\Psi_c^\top \Psi_c + n\alpha I_p)^{-1}\Psi_c^\top e_c.$$

For a singular value $s \ge 0$ of Ψ_c, the corresponding singular value of the coefficient operator is $s/(s^2 + n\alpha)$. The inequality $(s - \sqrt{n\alpha})^2 \ge 0$ implies

$$\frac{s}{s^2 + n\alpha} \le \frac{1}{2\sqrt{n\alpha}}.$$

The operator norm is therefore at most $1/(2\sqrt{n\alpha})$, giving $\|w\|_2 \le \|e_c\|_2/(2\sqrt{n\alpha}) = \sigma_e/(2\sqrt{\alpha})$. Finally, the fitted intercept gives $q(z) = \bar{e} + (\psi(z) - \bar{\psi})^\top w$. Cauchy–Schwarz and the triangle inequality yield

$$|q(z)| \le |\bar{e}| + (\|\psi(z)\|_2 + \|\bar{\psi}\|_2)\|w\|_2 \le |\bar{e}| + (R + \|\bar{\psi}\|_2)\frac{\sigma_e}{2\sqrt{\alpha}}.$$

Multiplication by $|\gamma|$ proves the correction bound. □

The mean-squared-error objective corresponds to a summed-squared-error objective with penalty $n\alpha$. Keeping the numerical value α unchanged when switching to a summed loss would define a different estimator. For multiple forecast horizons, the bound applies separately to each fitted residual vector and each selected pair (α_h, γ_h). This is a general property of ridge regression on bounded features, including the KAHM features considered here. It bounds the additive correction but does not bound the retained raw baseline or total forecast error. The experiment checks the feature, coefficient, and correction inequalities on every saved task and reports forecast errors separately.

Centered slope-only specialization. For Equation (26), let Φ_c have rows $(\varphi_m(x_i) - \bar{\varphi}_m)^\top$ and let $r_c = (r_{ih} - \bar{r}_h)_{i=1}^{n_A}$. Then

$$v_{m,h} = (\Phi_c^\top \Phi_c + n_A \alpha_{m,h} I)^{-1}\Phi_c^\top r_c.$$

The same singular-value inequality gives the coefficient bound in Equation (28). Cauchy–Schwarz applied directly to $(\varphi_m(x) - \bar{\varphi}_m)^\top v_{m,h}$ gives its correction bound, without a response-mean term. No intercept is reintroduced in this deployed slope. Finally, writing $e = Y - f_{\text{base}}(X)$ gives $(e - \eta\delta(X))^2 - e^2 = \eta^2\delta(X)^2 - 2\eta e\delta(X)$. Finite second moments make all terms integrable and justify taking the conditional expectation stated in Equation (29). This identity uses no independence assumption between the residual and the correction.

B Earlier residual forecasting readouts

The completed residual-forecast studies read out future observations through bounded features and retained baselines. This appendix preserves their transformations, fitting rules, and generic ridge bounds. Section 5 specifies the direct affine readouts for the G/P/K/R representation comparison, which separately transform membership and position blocks. Neither architecture estimates or iterates a transition matrix. Bounded decoder inputs are distinguished from the raw observable spaces analyzed in Section 4.

B.1 Joint common block and separate departure block

Each feature block is centered and scaled coordinatewise using A alone; feature standard deviations are floored at 10^{-6}. A tilde denotes this coordinatewise standardization. Apply the radial map T_R in Equation (11), with $R = \sqrt{p}$ for a block of width p. This radial map has norm at most R and is nonexpansive with respect to its argument. If the common and departure block dimensions are p_0 and p_d, respectively, set $R_0 = \sqrt{p_0}$ and $R_d = \sqrt{p_d}$. The bounded feature vectors are

$$\chi_K(z) = \mathrm{concat}\big(T_{R_0}(\widetilde{\psi}_0(z)), T_{R_d}(\widetilde{d}_K(z))\big), \qquad \chi_R(z) = \mathrm{concat}\big(T_{R_0}(\widetilde{\psi}_0(z)), T_{R_d}(\widetilde{d}_R(z))\big).$$

The two blocks are transformed separately and are not standardized again. Consequently, appending a departure does not rescale the common features. For J local models with stored ranks r_j, the dimensions are $p_0 = J + \sum_j r_j$ and $p_d = Jd$. The earlier dimension experiment requests partitions with 2, 4, and 8 models and at most 24 references per model. All fitted banks contain $J = 14$ models, so $p_d = 168$ for its 12-dimensional histories, whereas p_0 varies with the local ranks. Each candidate geometry is fitted without targets on A and remains fixed during selection and evaluation. All bounds use its actual dimensions.

B.2 Fitting and calibrating the correction

In the dimension study, let $f_{0,h}$ be the raw-history ridge baseline for horizon h. Given an unstandardized history x, let $\psi(x)$ denote a generic chosen bounded feature vector, such as χ_K or χ_R, after applying its A-fitted input scaler. The forecast is

$$\widehat{y}_{t+h} = f_{0,h}(x_t) + \gamma_h q_h(x_t), \qquad q_h(x) = b_h + \psi(x)^\top w_h. \tag{22}$$

Here $b_h \in \mathbb{R}$ is a free intercept, w_h is the coefficient vector, and γ_h controls the correction strength. For n_A fitting histories, the residual targets are $e_{ih} = y_{i,h} - f_{0,h}(x_i)$, where $y_{i,h}$ is the observed future at horizon h for fitting history i. The correction minimizes

$$\frac{1}{n_A}\sum_{i=1}^{n_A}\{e_{ih} - b_h - \psi(x_i)^\top w_h\}^2 + \alpha_h \|w_h\|_2^2.$$

When ψ contains both common and departure blocks, their coefficients are fitted jointly. Sharing the raw-history baseline and the common feature construction therefore does not fix the fitted common contribution across departure methods. Set B selects $\alpha_h \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$ by forecast error separately for each horizon and candidate cap. For each geometric procedure m, let $\widehat{y}^{(c)}_{m,h}$ be its B prediction after this penalty selection, with $\gamma_h = 1$. One cap is selected across horizons by

$$\widehat{c}_m = \operatorname*{arg\,min}_{c\in\{1,2,4,8,12,20\}} \frac{1}{|\mathcal{H}|}\sum_{h\in\mathcal{H}} \frac{\mathrm{MSE}_B(Y_h, \widehat{y}^{(c)}_{m,h})}{\max\{\mathrm{Var}_A(Y_h), 10^{-12}\}}. \tag{23}$$

Here Var_A is the empirical variance of the retained A targets with divisor n_A; exact ties select the smallest cap. The common, bounded KAHM, bounded RBF, and two unbounded-departure procedures each make their own choice. These choices are fixed before accessing C. Set C then selects $\gamma_h \in \{0, 1/4, 1/2, 1\}$ separately by horizon. Neither selection step refits the geometry or decoder, and set D is reserved for final evaluation. The cap objective uses uncalibrated B forecasts, so C does not enter dimension selection.

B.3 A uniform bound including the intercept

Define the fitting residual mean, centered root mean square, and feature mean as

$$\bar{e}_h = \frac{1}{n_A}\sum_i e_{ih}, \qquad \sigma_{e,h} = \left[\frac{1}{n_A}\sum_i (e_{ih} - \bar{e}_h)^2\right]^{1/2}, \qquad \bar{\psi} = \frac{1}{n_A}\sum_i \psi(x_i).$$

Proposition 5 (Centered ridge correction bound)**.** *For the mean-squared-error objective above with $n_A \geq 1$, $\alpha_h > 0$, and an unpenalized intercept, suppose that the feature map satisfies $\|\psi(x)\|_2 \leq R$ on its input domain. For any scalar γ_h, the fitted coefficients and correction satisfy*

$$\|w_h\|_2 \leq \frac{\sigma_{e,h}}{2\sqrt{\alpha_h}}, \qquad |\gamma_h q_h(x)| \leq |\gamma_h| \left[|\bar{e}_h| + (R + \|\bar{\psi}\|_2) \frac{\sigma_{e,h}}{2\sqrt{\alpha_h}} \right]. \tag{24}$$

For a concatenated bounded common and departure vector, $R = \sqrt{R_0^2 + R_d^2}$; for the common-only vector, $R = R_0$. For this decoder, the free intercept contributes the residual-mean term and cannot be omitted. This bound controls the correction, not the baseline, total forecast error, or sensitivity of the preceding reconstruction to changes in the history.

Proof idea. Eliminating the intercept gives $b_h = \bar{e}_h - \bar{\psi}^\top w_h$. If Ψ_c is the matrix whose rows are $(\psi(x_i) - \bar{\psi})^\top$ and e_c is the centered residual vector, then $w_h = (\Psi_c^\top \Psi_c + n_A \alpha_h I)^{-1} \Psi_c^\top e_c$. The coefficient operator has singular values $s/(s^2 + n_A \alpha_h) \leq 1/(2\sqrt{n_A \alpha_h})$. This yields the coefficient bound. Applying Cauchy–Schwarz to $q_h(x) = \bar{e}_h + (\psi(x) - \bar{\psi})^\top w_h$ yields the correction bound. Appendix A gives the complete argument, including the nonexpansiveness of the radial map.

The proposition applies separately to each selected horizon and is a property of bounded-feature ridge regression. A summed-squared-error objective would need penalty $n_A \alpha_h$ to define the same estimator. The radial transformation is nonexpansive in its own input, but this does not give a Lipschitz bound for the preceding signed reconstruction or for its fallback boundary.

Baseline recovery and predictive gain. The calibration grid includes $\gamma_h = 0$, which recovers the same fitted baseline exactly. This is an available forecast rather than a guarantee that a finite calibration set selects the best future forecast. The coefficient and correction bounds likewise make no claim about prediction accuracy. The experiment therefore evaluates error, numerical coverage, and the correction inequalities as separate outcomes.

B.4 Centered corrections to a frozen common predictor

The residual-target study fixes a separately fitted common forecast $f_{\mathrm{com},h}$ and a shared bias before comparing departures:

$$f_{\mathrm{base},h}(x) = f_{\mathrm{com},h}(x) + \beta_h, \qquad \beta_h = \frac{1}{n_B} \sum_{i \in B} \{ y_{i,h} - f_{\mathrm{com},h}(x_i) \}. \tag{25}$$

The common forecast is fitted on all of A and selected on B. Its coefficients and the B-fitted bias are then shared across the departure comparisons and frozen before C. Thus the common component cannot compensate for a change of departure coordinates during residual fitting.

Let $\varphi_m(x)$ be the separately standardized and radially bounded departure block for method m, with $\|\varphi_m(x)\|_2 \leq R_{d,m}$, and let $\bar{\varphi}_m$ be its A mean. For either in-sample or out-of-fold residual targets r_{ih}, write $\bar{r}_h = n_A^{-1} \sum_i r_{ih}$ and fit

$$v_{m,h} = \arg\min_v \left[\frac{1}{n_A} \sum_i \{ r_{ih} - \bar{r}_h - (\varphi_m(x_i) - \bar{\varphi}_m)^\top v \}^2 + \alpha_{m,h} \|v\|_2^2 \right], \qquad \alpha_{m,h} > 0. \tag{26}$$

In-sample targets use $r_{ih} = y_{i,h} - f_{\mathrm{com},h}(x_i)$; out-of-fold targets use the common forecast fitted without the fold containing the episode of x_i. The deployed forecast is

$$\widehat{f}_{m,h}(x) = f_{\mathrm{base},h}(x) + \eta_{m,h} \delta_{m,h}(x), \qquad \delta_{m,h}(x) = (\varphi_m(x) - \bar{\varphi}_m)^\top v_{m,h}. \tag{27}$$

It contains no additional residual-mean intercept. Subtracting the constant shared bias before centering the fitting targets would leave the slope unchanged. The correction has zero empirical mean on A; its mean on another distribution need not vanish. Setting $\eta_{m,h} = 0$ recovers the same shared baseline for every departure method.

Bound for the deployed slope. Define $\sigma_{r,h}^2 = n_A^{-1}\sum_i (r_{ih} - \bar{r}_h)^2$. The spectral argument of Proposition 5 immediately gives

$$\|v_{m,h}\|_2 \le \frac{\sigma_{r,h}}{2\sqrt{\alpha_{m,h}}}, \qquad |\eta_{m,h}\delta_{m,h}(x)| \le |\eta_{m,h}|(R_{d,m} + \|\bar{\varphi}_m\|_2)\frac{\sigma_{r,h}}{2\sqrt{\alpha_{m,h}}}. \tag{28}$$

The residual-mean term is absent because it is absent from the deployed correction. The fixed common forecast and shared bias remain outside this bound. This specialization is generic bounded-feature ridge algebra and applies equally to the matched RBF departure.

Residual alignment and predictive gain. Condition on all fitted and selected quantities, and take expectations under a specified evaluation distribution for (X, Y). Suppress the horizon and method indices, and suppose Y, $f_{\text{base}}(X)$, and $\delta(X)$ have finite second moments. For fixed η, define $\mathcal{R}(f) = \mathbb{E}[(Y - f(X))^2]$. Expansion of the square gives the elementary identity

$$\mathcal{R}(f_{\text{base}} + \eta\delta) - \mathcal{R}(f_{\text{base}}) = \eta^2\mathbb{E}[\delta(X)^2] - 2\eta\mathbb{E}[(Y - f_{\text{base}}(X))\delta(X)]. \tag{29}$$

For $\eta > 0$, improvement holds exactly when $\mathbb{E}[(Y - f_{\text{base}}(X))\delta(X)] > \eta\mathbb{E}[\delta(X)^2]/2$. A bound on correction magnitude does not establish that alignment. The identity concerns errors of the deployed full-A baseline even when fitting uses fold-model residuals; it does not equate those two residual distributions or identify the cause of an observed performance difference.

C Complete reconstruction-departure evaluation

This appendix specifies the observation laws, fitting procedure, complete comparison set, stability checks, and resource measurements for Section 7. The reproduction materials contain the implementation, saved predictions and model states, and machine-readable analyses under `extensions/dimension_forecasting`. All tables and figures in this appendix are generated from those analyses.

C.1 Observation laws and data roles

The linear negative control is

$$y_t = 0.65y_{t-1} - 0.15y_{t-2} + \epsilon_t, \qquad \epsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, 0.1^2),$$

where $\mathcal{N}(\mu, \sigma^2)$ denotes a normal distribution with mean μ and variance σ^2. The process starts with two zeros; 256 stored values are discarded, and 72 values beginning at index 256 are retained. The nonlinear autoregression is

$$y_t = \tanh\{1.25y_{t-1} - 0.65y_{t-2} + 0.25\sin(3y_{t-1}) + u_t\}, \qquad u_t \overset{\text{iid}}{\sim} U[-0.1, 0.1],$$

where $U[a, b]$ denotes the uniform distribution on $[a, b]$. Its two initial values are independent draws from $U[-0.8, 0.8]$, and 80 stored values are discarded before retaining an episode.

The Van der Pol dynamics are $\dot{x} = v$ and $\dot{v} = (1 - x^2)v - x$. The Duffing dynamics are $\dot{x} = v$ and $\dot{v} = -0.2v + x - x^3 + 0.3\cos(1.2t + \varphi)$. Here x is position, v is velocity, a dot denotes differentiation with respect to continuous time t, and the Duffing phase φ is uniform on $[0, 2\pi]$. Initial position and velocity are independent draws from $U[-2, 2]$. The ordinary differential equations are integrated in float64 arithmetic using the classical fourth-order Runge–Kutta method with step 0.025, observation interval 0.1, and a discarded initial time interval of length ten. The scalar observation is $0.95\tanh(x) + \xi$, with independent sensor noise $\xi \sim U[-0.02, 0.02]$. The resulting numerical integration and observation map define the tested simulation law; the observations are not exact continuous-time trajectories.

For each law and repetition, the episode sets A, B, C, and D are generated once. Additional noise has the form $\eta s_A Z$, where $\eta \in \{0, 0.1\}$, $Z \sim \mathcal{N}(0, 1)$ independently at each stored observation, and s_A is the empirical standard deviation, with divisor equal to the number of observations, of all A episode values before this

addition. The underlying episodes and Z arrays are paired across the two noise levels. Added noise affects both histories and observed future targets. At $\eta = 0$, the oscillator sensor noise is still present. The B/C/D linear observations are Gaussian conditional on the A-estimated noise amplitude, while their marginal law across repetitions is a variance mixture. Finite burn-in does not imply exact stationarity, and the added Gaussian noise means that the observed responses are not uniformly bounded.

The master seed is 2026110103 and repetitions are indexed by $r = 0, \ldots, 23$. A uint32 stream seed is the first output of NumPy's `SeedSequence` applied to an integer tuple. Law indices follow linear AR, nonlinear AR, Van der Pol, and Duffing; set indices are A/B/C/D = 0/1/2/3. Base, noise, and retained-row streams use $(2026110103, r, \text{law}, \text{set}, \text{purpose})$, with purposes 1001, 1002, and 1003. Fitting uses $(2026110103, r, \text{law}, 1004)$, and the repetition identifier uses $(2026110103, r, 1000)$. The 1,272 intended-distinct stream identifiers are checked for collisions and for overlap with the 2,173 distinct identifiers recorded for earlier principal and dimension studies. Base episodes and standardized noise arrays for different sets use independent streams. The additional-noise amplitude is estimated on A and shared across sets. Fitting seeds, row choices, references, and noise draws are paired across caps and noise levels. Using zero-based indices, the 13 origins $11, 15, \ldots, 59$ each have a length-12 history ending at the origin and targets at offsets 1, 4, and 12. A/B/C/D contain 64/32/32/128 episodes. The first three sets retain 512/256/256 histories; D retains all 1,664 histories. This gives 958,464 scored target values per method over the full experiment. Inference nevertheless uses 24 independent repetition aggregates.

A fits every scaler, reference bank, kernel factor, baseline, and decoder. B selects penalties and raw-model settings separately by horizon, then one dimension cap across horizons for each geometric procedure. Its cap objective uses the uncalibrated predictions in Equation (23); all five cap choices are saved before C is loaded. C selects each candidate's mixing coefficients and then one complete non-KAHM procedure within each comparison context. All fitted states and the complete task roster are sealed before D episodes are generated. D histories are supplied to fixed predictors; their targets remain separate until every prediction is sealed. A later origin can legitimately contain observations that were targets for an earlier origin in the same episode, but it causes no fitting or cross-query adaptation. All uncertainty calculations use independent repetitions, not the overlapping origins.

C.2 Reference geometry and numerical evaluation

The geometric bank first standardizes each history coordinate using its A mean and standard deviation. A standard deviation at most 10^{-12} is replaced by one. At most 512 A rows enter the bank fit. Separate k-means partitions request 2, 4, and 8 groups, using five initializations and at most 150 iterations. If there are too few distinct input rows, the requested group counts are reduced to the available number. An unresolved empty-group configuration is retried with halved group counts. Within each group, at most 24 reference rows are sampled without replacement. The references are stored in float32. The native PCA converts its input to float64 and uses the centered sample covariance. Native candidate cap $c \in \{1, 2, 4, 8, 12, 20\}$ initializes the requested rank at $\min\{c, d, n_j - 1\}$. Eigenvalues must exceed double-precision machine epsilon. The requested rank is decremented and the native PCA recomputed if the smallest range of a projected reference coordinate is below 10^{-3}, stopping when the range condition holds or requested rank one is reached. A rank-one model may therefore remain below that threshold. The explicit fixed-cap-two comparison instead initializes $\min\{2, d - 1, n_j - 1\}$ and uses the same subsequent numerical rules. The actual stored rank is the number of columns of P_j, not the supplied cap. The forecasting wrapper preserves its explicit zero-rank affine fallback for singleton or constant references; this is not claimed as successful native PCA. All cap requests, initial ranks, stored ranks, reference counts, and reduction or fallback reasons are recorded.

For the row-stored references Q_j, define $\mu_j = n_j^{-1} \sum_{i=1}^{n_j} q_{ji}$ and

$$s_j = \max\left\{ \frac{1}{n_j} \sum_{i=1}^{n_j} \|q_{ji} - \mu_j\|_2^2, 10^{-6} \right\}.$$

This is the scale used in the common locality weights. The projected sample covariance is

$$S_j = \frac{1}{n_j - 1} \sum_{i=1}^{n_j} P_j^\top (q_{ji} - \mu_j)(q_{ji} - \mu_j)^\top P_j.$$

Its inverse defines the shared Gaussian kernel in Section 3. Kernels and references are identical between KAHM and positive-RBF reconstruction within a bank. Independent cap choices can select different projected kernels and bases.

The KAHM regularizer is fitted from reference reconstruction, without forecast targets. Suppress the local index j, let $Q \in \mathbb{R}^{n \times d}$ contain the local references, and define the reference energy $\nu = \|Q\|_F^2/(nd)$, where $\|\cdot\|_F$ is the square root of the sum of squared matrix entries. Initialize $\lambda^{(0)} = 2.5\nu$. At iteration a, compute

$$C^{(a)} = (K + \lambda^{(a)} I_n)^{-1}, \qquad \lambda^{(a+1)} = 2\nu + \frac{\|Q - KC^{(a)}Q\|_F^2}{nd}.$$

The iteration stops when $|\lambda^{(a+1)} - \lambda^{(a)}| \le 10^{-3}$ or after 100 inverse computations. The reconstruction uses the last computed $C^{(a)}$ and its corresponding $\lambda^{(a)}$, rather than the subsequent proposal $\lambda^{(a+1)}$. An unexpected factorization failure or invalid regularizer stops the experiment; it is not converted into a successful candidate. This rule determines a reference-based regularizer; it does not optimize forecasting loss.

For each query, the maximum log-kernel entry is subtracted before exponentiation. The resulting common positive rescaling leaves both normalized reconstruction ratios unchanged. Signed cancellation is checked using ρ_j in Equation (6). The affine fallback makes the corresponding raw departure zero if the ratio cannot be evaluated reliably. The experimental results below report how often this occurs.

C.3 Decoders, comparison grids, and descriptive losses

The common and departure feature columns are standardized separately using A, with each standard deviation replaced by its maximum with 10^{-6}. The blocks are then radially bounded as specified in Equation (11), with no subsequent scaling. The two unbounded-departure variants retain the bounded common block and omit only the departure's radial transformation.

Every ridge head uses mean squared error plus $\alpha\|w\|_2^2$ and an unpenalized intercept, with $\alpha \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$. Equivalently, the unnormalized sum-of-squares objective uses penalty $n\alpha$ for n fitting rows. Raw-history ridge and RBF methods standardize their input coordinates on A with a 10^{-6} scale floor. The raw RBF method concatenates standardized histories with at most 64 Nyström features, then standardizes the concatenated columns using A. The squared Gaussian bandwidth equals the median of all strictly positive squared landmark distances multiplied by 0.5, 1, or 2, with no bandwidth floor. When there are no strictly positive landmark distances, the median used by this rule is set to one. B selects this factor and the ridge penalty separately for each horizon. For a landmark Gram matrix whose largest eigenvalue is $\zeta_{\max}$, the Nyström transform retains eigenvalues above $\max\{10^{-12}, 10^{-10}\zeta_{\max}\}$. Let K_{MM} be the symmetrized landmark Gram matrix. Let the columns of V be its retained orthonormal eigenvectors, and let the diagonal matrix Λ contain their positive eigenvalues. If $k_M(u)$ is the vector of Gaussian similarities between a standardized query u and the M landmarks, its Nyström coordinates are $\Lambda^{-1/2}V^\top k_M(u)$. The Gaussian is $\exp\{-\|u - u'\|_2^2/(2\ell^2)\}$, with squared bandwidth ℓ^2 chosen by the median-distance rule above. These coordinates are concatenated with the standardized raw history before column scaling and ridge fitting.

Histogram gradient boosting uses 100 iterations, L2 regularization one, and no early stopping. B selects learning rate 0.05 or 0.1 and maximum leaf count 7 or 15 separately by horizon. Persistence predicts the latest observed history value at every horizon. Every non-ridge candidate is then blended with ridge using a C-selected coefficient from $\{0, 1/4, 1/2, 1\}$. Ties are resolved by the recorded candidate order. The non-KAHM policy chooses in this order among ridge, raw RBF, boosting, persistence, the common basis, and the two RBF-departure variants; both KAHM variants are excluded. Table 5 reports all methods and conditions.

The machine-readable files `horizon_losses.csv` and `condition_horizon_summary.csv` retain every method, law, noise level and horizon, including raw MSE and the A-only loss scale. They are not pooled with historical

Table 5: All ten procedures, every declared law and both added-noise levels. Entries are mean horizon-averaged A-variance-normalized D MSE over 24 repetitions; lower is better. The first six columns enter the nonlinear aggregate with equal weight. The final two form the separate linear control. These conditionwise summaries are descriptive. "Unbounded" omits only the departure radial bound. Persistence and raw nonlinear controls include their C-selected blend with ridge.

Procedure	Nonlinear AR		Van der Pol		Duffing		Linear AR	
	$\eta = 0$	0.1	$\eta = 0$	0.1	$\eta = 0$	0.1	$\eta = 0$	0.1
Raw ridge	0.02200	0.03451	0.07470	0.13584	0.01955	0.06873	0.90079	0.90253
Raw RBF	0.01996	0.03243	0.01326	0.06486	0.01296	0.04511	0.90383	0.90556
Boosted trees	0.02038	0.03231	0.00380	0.05863	0.01351	0.04908	0.90422	0.90473
Persistence	0.02205	0.03451	0.07470	0.13588	0.01975	0.06955	0.90140	0.90382
Common features	0.01928	0.03115	0.02126	0.05902	0.01406	0.04587	0.90139	0.90319
Bounded KAHM departure	0.01928	0.03109	0.00935	0.05735	0.01147	0.04494	0.90230	0.90454
Bounded RBF departure	0.01930	0.03119	0.01064	0.05640	0.01289	0.04431	0.90198	0.90366
Unbounded KAHM departure	0.01949	0.03149	0.01013	0.05929	0.01097	0.04511	0.90237	0.90459
Unbounded RBF departure	0.01937	0.03132	0.01104	0.05861	0.01157	0.04471	0.90244	0.90385
Selected non-KAHM	0.01923	0.03110	0.00380	0.05615	0.01175	0.04463	0.90600	0.90700

losses that use different normalizations. The `primary_repetition_contrasts.csv` file contains all 24 margin-gap values for each primary contrast, allowing the six lower bounds to be recomputed without importing any forecasting model.

Table 6: Nonlinear horizon-specific SMSE, averaged equally over the six law–noise conditions and 24 repetitions. Except for the first two fixed KAHM rows, procedures use their independently selected caps where applicable. Every non-ridge forecast includes its C calibration. These descriptive summaries retain the original A-only variance scales.

Procedure	Horizon 1	Horizon 4	Horizon 12
KAHM, fixed cap 2	0.011654	0.021471	0.056529
KAHM, native cap 20	0.011207	0.019108	0.066862
Raw ridge	0.013041	0.036954	0.127674
Raw RBF	0.011248	0.019724	0.063314
Boosted trees	0.011742	0.020626	0.056490
Persistence	0.013056	0.036971	0.128193
Common features	0.011149	0.018970	0.065207
Bounded KAHM departure	0.011201	0.019491	0.056043
Bounded RBF departure	0.011204	0.019516	0.056647
Unbounded KAHM departure	0.011378	0.020254	0.056611
Unbounded RBF departure	0.011226	0.019790	0.057292
Selected non-KAHM	0.011281	0.019431	0.052623

C.4 Dimension selection and realized geometry

Table 7 reports all five geometric procedures' dimension choices. Native caps twelve and twenty have identical arrays and forecasts at ambient dimension twelve, and exact tie resolution selects twelve. Native cap two and the fixed-cap-two policy are also identical. These identities are checked on bases, features, fitted heads, B choices, C calibration and final predictions; the policy metadata remain distinct. Raw ridge, raw RBF, boosting, and persistence are fitted once per task and shared across the three contexts. The cap-dependent heads and seven banks are fitted separately without outcome-dependent deduplication.

There are 192 banks and 2,688 local models per cap rule. Local reference counts range from 2 to 24; 2,670 local reference sets per rule contain 24 rows. Table 8 reports stored ranks and feature widths. For native cap twenty, 2,685 local models have rank twelve and the remaining three have two references and rank one. All banks have 14 local models and a stacked principal-basis rank of twelve. No additional positive-eigenvalue or projected-range reduction occurs in this sample. The observed ranks describe these banks, not a guarantee for arbitrary datasets.

Table 7: B-selected cap counts. Each procedure contributes 144 nonlinear and 48 linear tasks; it selects one cap across all three horizons before C calibration. Exact cap-twelve/cap-twenty ties explain the zero final column. The common and RBF rows use their own choices, not KAHM's choices.

Procedure	Panel	1	2	4	8	12	20
Common features	Nonlinear	13	10	68	40	13	0
Common features	Linear	35	9	3	0	1	0
Bounded KAHM departure	Nonlinear	15	22	68	31	8	0
Bounded KAHM departure	Linear	4	14	18	9	3	0
Bounded RBF departure	Nonlinear	26	29	58	26	5	0
Bounded RBF departure	Linear	14	24	3	5	2	0
Unbounded KAHM departure	Nonlinear	17	27	67	25	8	0
Unbounded KAHM departure	Linear	2	5	27	11	3	0
Unbounded RBF departure	Nonlinear	19	39	59	20	7	0
Unbounded RBF departure	Linear	9	26	10	2	1	0

Table 8: Realized geometry over the complete task roster. Rank statistics refer to the 2,688 local models in each row; "Rank 12" counts full-ambient local bases. Width ranges refer to 192 complete banks. The full departure readout adds the common and departure widths. Stacked-basis rank is twelve for every bank, even when its individual local ranks are small.

Bank rule	Rank range	Mean rank	Rank 12	Common width	Departure width
Fixed cap 2	1–2	1.999	0	40–42	168
Native cap 1	1–1	1.000	0	28–28	168
Native cap 2	1–2	1.999	0	40–42	168
Native cap 4	1–4	3.997	0	64–70	168
Native cap 8	1–8	7.992	0	112–126	168
Native cap 12	1–12	11.988	2685	160–182	168
Native cap 20	1–12	11.988	2685	160–182	168

The common and departure contributions in Section 7 use the KAHM-selected bank, not the independently selected common procedure. The matched common mean loss is 0.032702, compared with 0.031775 for independently selected common features. Table 9 reports the exact mean loss decomposition relative to fixed cap two. For native cap twenty, the incremental departure benefit is only −0.000653, compared with −0.004967 at cap two. A change in complete KAHM accuracy must therefore be distinguished from a change in the incremental value of its reconstruction departure.

Table 9: Nonlinear loss decomposition relative to the fixed-cap-two procedure. Total change equals common change plus increment change. The final column gives the new KAHM-minus-common loss on the same bank. Negative values in the first two columns denote improvement; a positive increment change denotes a weaker departure benefit. These are descriptive identities between calibrated predictor losses.

Bank rule	Total change	Common change	Increment change	$L_K - L_0$
KAHM-selected bank	-0.000973	-0.002150	0.001177	-0.003790
Native cap 20	0.002507	-0.001806	0.004313	-0.000653

C.5 Stability, numerical safeguards and correction activation

The bounded-feature checks cover 192 tasks, seven banks, all four data stages, and each of the three bounded procedures. Coefficient checks cover all five geometric heads and three horizons for each task and bank; global correction checks apply to the three bounded heads. The independent analyzer checks the recorded fitted-state bound formulas and recomputes actual D corrections as the saved forecast minus the saved raw ridge forecast. It does not claim to reconstruct the fitted coefficient matrices from summary statistics alone. Table 10 reports maxima over all seven banks and the complete roster; maxima in different columns need not come from the same task or horizon.

Table 10: Observed ratios to the deterministic bounded-feature and ridge-readout limits. Feature ratios concern each procedure's complete feature vector and its actual radius, including both blocks for the two appended procedures. A zero correction with a zero bound contributes ratio zero. "Violations" sums failed feature, coefficient and correction checks at the fixed numerical tolerances. These are implementation checks, not confidence intervals for forecast risk.

Procedure	Feature ratio	Coefficient ratio	Correction ratio	Violations
Common features	0.9935	0.7154	0.2824	0
Bounded KAHM departure	0.9983	0.6735	0.2207	0
Bounded RBF departure	0.9974	0.6738	0.2347	0

The maximum taskwise ratio of KAHM maximum absolute error to ridge maximum absolute error is 1.4673, and no task fails a declared stability screen. All 5,376 task–bank–stage diagnostic records report zero affected rows for coordinate, gate, kernel, KAHM, and RBF query safeguards. No local model uses a rank-zero fallback, and no unexpected factor failure occurs. Counts refer to unions over local models within a task, bank and stage; the same query would count again if a different bank or stage invoked a guard. No method fails completely and no final prediction cell requires recovery. These checks cover every specified task and cap candidate.

C.6 Measured costs and reproducibility limits

Each task runs in a fresh process with one assigned CPU and one numerical thread; up to three tasks run concurrently on distinct CPUs. The charged primitive-node costs exclude diagnostic-only rank calculations; full-suite timings, including diagnostics, are retained separately. Node accounting includes model candidates that are actually evaluated, rather than only the winning hyperparameters. Wall and CPU times depend on the recorded machine and software environment. The peak whole-task process RSS is 162.9 MiB; it includes imports, all candidates, shared diagnostics and evidence arrays, and is not a per-method RAM estimate. Serialized state and owned array buffers omit transient workspace and do not measure online

Table 11: C-selected correction activation. Nonlinear and linear counts have 432 and 144 task/horizon cells, respectively. The four mixing-weight columns combine all 576 cells. A policy choice of raw ridge is counted as zero correction; raw ridge itself has no mixing selection. Nonzero mixing is not an empirical guarantee of D improvement, and no minimum activation frequency was part of the decision.

Procedure	NL active	Linear active	$\gamma = 0$	1/4	1/2	1
Raw RBF	432	56	88	53	147	288
Boosted trees	416	43	117	104	189	166
Persistence	52	20	504	71	1	0
Common features	431	51	94	14	52	416
Bounded KAHM departure	431	52	93	21	73	389
Bounded RBF departure	430	53	93	21	62	400
Unbounded KAHM departure	432	51	93	26	98	359
Unbounded RBF departure	432	48	96	23	68	389
Selected non-KAHM	431	72	73	38	99	366

Table 12: Whole-procedure choices made by the C-selected non-KAHM policy. Each task contributes one choice across all horizons after each candidate's own horizon-wise C mixing selection. Both KAHM arms are excluded. These are validation choices, not choices made from D errors.

Law	η	Ridge	Raw RBF	Boosting	Persist.	Common	RBF	RBF unb.
Nonlinear AR	0.0	0	0	0	0	8	3	13
Nonlinear AR	0.1	0	0	0	0	6	6	12
Van der Pol	0.0	0	0	24	0	0	0	0
Van der Pol	0.1	0	0	7	0	0	16	1
Duffing	0.0	0	2	1	0	0	2	19
Duffing	0.1	0	8	2	0	3	6	5
Linear AR	0.0	0	7	5	5	1	3	3
Linear AR	0.1	0	6	6	4	1	1	6

Table 13: Mean charged resources over all 192 equal-sized task suites. Fitting and selection include required A fitting, B selection and C calibration. Dimension-selected procedures pay for all six native cap candidates. Shared computation nodes are charged once to every procedure requiring them; policy fitting pays the union of its non-KAHM candidates. State denotes the pruned selected deployment object, while arrays count its owned NumPy buffers. The final two rows give fixed-cap KAHM controls. These are descriptive implementation costs, not isolated speed benchmarks or minimum deployment complexity.

Procedure	Wall (s)	CPU (s)	State (KiB)	Arrays (KiB)
Raw ridge	0.001	0.001	1.1	0.6
Raw RBF	0.015	0.015	68.1	66.7
Boosted trees	0.681	0.681	370.5	332.1
Persistence	0.001	0.001	1.2	0.7
Common features	0.187	0.187	17.3	11.7
Bounded KAHM departure	0.490	0.490	136.9	129.5
Bounded RBF departure	0.380	0.380	68.7	61.9
Unbounded KAHM departure	0.484	0.484	136.7	129.2
Unbounded RBF departure	0.377	0.377	68.2	61.3
Selected non-KAHM	1.261	1.261	136.9	123.7
KAHM, fixed cap 2	0.085	0.086	122.1	114.7
KAHM, native cap 20	0.101	0.101	183.3	175.8

update latency. The complete fit stage takes 308.2 seconds and the prediction stage 227.5 seconds on this shared host, including worker and evidence overhead. They are not per-method latency estimates. The study imposes no application-specific cost ceiling.

The analysis program verifies the frozen source and staged input checksums, the complete task roster, exact target correspondence, A variance scales, policy predictions, nonfinite/failure accounting and all primary aggregation. An independent post-release audit regenerates episodes from the pinned generators and independently checks seeds, noise, windows, targets, B and C choices, local bases, and all 27 numerical tables. The manuscript builder verifies the analysis table hashes before producing these displays.

The frozen source inventory has 151 files. Model states are hashed before and after prediction, all target-bearing files have explicit release checks, and every candidate forecast is retained. Six disjoint numerical archives group repetitions 0–3, 4–7, 8–11, 12–15, 16–19, and 20–23; the first also contains global inventories and seals. Their manifest records every member checksum. The archived forecast arrays define the reported scores. Re-evaluating serialized states additionally requires compatible numerical kernels and package versions; processor-dependent vectorized arithmetic can change individual floating-point values.

Numerical development checks, model replay and independent calculations are consistency checks; they do not constitute a new scientific replication. The protocol was specified locally before evaluation and was informed by the earlier experiments. It was not publicly preregistered. Neither these known-law repetitions nor the algebraic distinctness certificate establishes a natural-application forecasting claim.